

Defending Digital Democracy: Mitigating Redundancy Attacks in Voting Advice Applications

Liv Weiss Damien Berriaud Frederic Berdoz Roger Wattenhofer
ETH Zürich, Zürich, Switzerland liweiss, dberriaud, fberdoz, wattenhofer@ethz.ch

Abstract—Voting Advice Applications (VAAs) match voters with candidates based on policy preferences, but their default recommendation algorithms inherently assume that all survey questions are equally important. Malicious actors can exploit this vulnerability by injecting redundant or correlated questions to artificially amplify specific policy dimensions and skew recommendations. We propose an automated defense mechanism that mitigates adversarial redundancy by assigning non-uniform weights to survey questions in the standard distance-based recommendation algorithm. These weights are derived from the recently introduced clone-robust weighting (CRW) axiomatic framework, which operates on an embedding of questions in a relevant metric space, thereby reducing the influence of redundant or strategically injected questions. The framework is instantiated using transformer-based sentence embeddings of survey questions and evaluated using data from the 2023 Swiss National Council elections on SmartVote. A systematic comparison across off-the-shelves embedding models and parameter settings shows that combining E5-Instruct with CRW improves robustness to artificially crafted clones in 98.7% of question-clone pairs, and offsets 19 to 62% of artificial gains in party visibility under targeted preferential attacks. These results highlight the effectiveness of the proposed mechanism as an automated first-line defense, and open the door to more ambitious instantiations that target response pattern similarity—an arguably broader notion of redundancy.

Index Terms—Voting Advice Application, Clone-robust Weighting, Semantic Redundancy, Sentence Embedding Models, SmartVote Dataset.

I. INTRODUCTION

Voting Advice Applications (VAAs) are online platforms that match voters with political candidates based on their responses to policy questionnaires. Deployed in over 30 countries [23], VAAs have become an integral part of modern democratic infrastructure, often being consulted by up to 50% of voters [23] with documented effects on both voter turnout and vote choice [16]. In Switzerland, the VAA platform SmartVote was consulted by up to 20% of eligible voters during the 2023 National Council elections [19]. Its influence extends beyond mere consultation: 67% of SmartVote users reported that the platform had influenced their voting behavior [11], with 15% stating they had fully adopted the given recommendations [10]. When a single platform shapes the information environment for a fifth of the electorate, the integrity of its recommendation algorithm is a matter of direct democratic consequence.

Most VAAs, including SmartVote, push voters to elicit their preferences by selecting private weights for each question. In practice, however, this input is often coarse, and a majority of

users do not provide any weights at all, c.f. Appendix VIII-E. In such cases, the platform defaults to computing voter-candidate distances using metrics that weight all questions equally [2]. This creates a structural vulnerability: if multiple questions in the questionnaire measure the same underlying policy dimension, whether by accident or by design, that dimension receives disproportionate influence over the final recommendations. An adversarial question designer can exploit this by injecting paraphrased or topically redundant questions, effectively amplifying the weight of a favored policy issue without any individual question appearing suspicious.

While this vulnerability was already identified in previous work [2], its full impact and potential mitigation were left unexplored. Recent advances in computational social choice provide a principled approach through the Clone-Robust Weighting (CRW) framework [3, 4]. This framework addresses how to define prior weights in the presence of redundancy, assuming similarity can be captured by a metric. It prescribes three principles: (i) symmetry, requiring weights to respect the structure of the space; (ii) continuity, ensuring stability under small perturbations either at the element or at the set level; and (iii) clone-robustness, meaning that adding near-duplicates should not distort the global distribution, but only redistribute weight locally. In effect, CRW identifies weighting functions that assign weights inversely proportional to the local density, so that a tightly-clustered elements collectively receive similar total weight to a single isolated element.

The effectiveness of CRW, however, depends entirely on the choice of embedding space used to measure similarity. This choice becomes critical when applying CRW to the weighting of political questions in VAAs: should the embedding space reflect a notion of political distance grounded in population responses—thereby capturing culturally specific structures, but potentially obscuring minority viewpoints? Or should it instead aim to minimize such population bias, relying solely on semantic similarity between questions? Yet an additional complication arises from question negation in the VAA setting: pairs such as “Should taxes be increased?” and “Should taxes be restricted?” are semantically opposite, yet encode the same policy dimension. Any defense that fails to treat them as redundant leaves a substantial attack surface open.

Contribution: This work provides the first systematic empirical validation of defenses against redundancy attacks in questionnaire-driven recommendation systems such as VAAs. Our contribution is threefold. First, we formalize question

redundancy in VAAs and characterize the structural vulnerability it induces, showing that even minimal redundancy—such as the injection of a single approximate clone—leads to immediate and measurable distortions in voter–candidate recommendations, with effects that scale predictably with the adversarial effort. Second, we introduce a general defense framework for VAAs based on clone-robust weighting (CRW). Building on prior theoretical work, we adapt CRW to the questionnaire setting and clarify its design space, highlighting how different choices of representation, distance, and interaction radius determine which notions of redundancy are captured and corrected. Third, we develop a comprehensive experimental pipeline based on synthetically generated clones spanning a wide spectrum of semantic difficulty and attack configurations, and use it to evaluate a semantic-based instantiation of our defensive framework. Our results show that CRW-based defenses consistently improve robustness across models, attack settings, questions and behavioral noise mitigating 17-47% of induced distortions. At the party level, they substantially reduce targeted visibility manipulation, decreasing artificial gains by up to 62% under minimal attacks.

Full version: The full paper (available here) provides: (1) additional background on the Swiss political system and SmartVote, including the full questionnaire and data preprocessing steps; (2) extended methodological information, covering clone generation, embedding model instructions, and procedures for model and hyperparameter selection; (3) detailed results on the agreement between robustness measures, per-question vulnerability, robustness across clone types, the stability of hyperparameter selection over the full questionnaire, and the unmitigated impact of targeted party manipulation; (4) complementary analyses on predictors of question vulnerability and the effectiveness of embedding-based mitigation strategies; and (5) the code used to run all experiments, enabling easy reproduction of the numerical results given access to the dataset.

II. RELATED WORK

This work lies at the intersection of three research areas.

a) Voting Advice Applications: A recurring finding in VAA research is that recommendations are highly sensitive to design choices. Louwerse and Rosema [14] demonstrate that switching the distance metric (e.g., from ℓ_2 to ℓ_1) substantially changes which candidates are recommended, even on the same data. Walgrave et al. [26] show that question selection alone can shift recommendations, while Isotalo [8] identifies transparency and interactivity issues in Finnish VAAs. These findings establish that VAA recommendations depend on algorithmic and design choices, not only on the underlying preference data. Berdoz et al. [2] analyze adversarial robustness in VAAs, identifying 11 manipulation strategies exploitable by candidates, platform operators, and question designers using SmartVote data. While some candidate-level attacks yield larger immediate visibility gains, the questionnaire-level vulnerability of *question correlation*—implicit weight amplification through redundant questions—is arguably more

insidious. Unlike candidate-level attacks, which require misrepresentation by individual candidates, this vulnerability can be exploited by a single question designer and affects all voter–candidate pairs simultaneously. Berdoz et al. [2] identify the threat but leave its quantification and mitigation open; we address both by measuring the full distortion from question cloning and deploying CRW as an automated countermeasure.

b) Clone-Robust Aggregation: The problem of redundancy distorting collective outcomes has deep roots in social choice theory. Tideman [24] established the independence of clones criterion for voting rules, while mechanism design addresses similar concerns through false-name-proofness [6]. Closer to our setting, clone-robustness was identified as a key property for the evaluation of models across diverse benchmarks in machine learning [1, 15]. While these works move the discussion toward judgment aggregation, they—like the classical literature—primarily focus on the impact of exact copies, for the lack of a rigorous axiomatic treatment of approximate clones. The challenge of handling these near-identical entities has only recently been addressed. Procaccia et al. [20] made initial progress in preference learning and RLHF by using Voronoi-region weighting to discount clustered alternatives. However, the first principled and axiomatic treatment of approximate clones was introduced by Berriaud and Wattenhofer [3]. Their framework for clone-robust weighting satisfies three key axioms: symmetry, continuity, and locality. Initially developed for Euclidean spaces, this framework was later generalized to arbitrary metric spaces [4], providing the theoretical tools to handle non-Euclidean data. We apply this generalized framework to mitigate adversarial question redundancy in VAAs, constituting, to our knowledge, the first real-world application of CRW to political data and recommendation pipelines.

c) Sentence Embedding Models: To instantiate CRW, we require a distance metric over questions, which we obtain using transformer-based sentence embedding models [25, 7]. SBERT [21] fine-tunes BERT with contrastive learning on semantic textual similarity (STS) data, producing fixed-size sentence embeddings comparable via cosine similarity. SBERT and its descendants form the *STS model family* evaluated in this paper: models trained to place sentences of similar *meaning* close together in embedding space. Lee et al. [12] demonstrate that domain-specific fine-tuning allows SBERT to mirror the correlation structure of political beliefs, suggesting that embedding models can be adapted to capture latent ideological dimensions directly. A second family relevant to our work are *text retrieval* models, trained to identify topically relevant documents rather than to judge semantic equivalence. Models such as E5 [27], Jina-v3 [22], GTE [13], and Qwen3 [29] fall into this category. The distinction matters for CRW because the two training objectives produce different behavior on negated pairs: STS models assign low similarity to semantically opposite sentences, though lexical overlap keeps scores moderately high in practice, while retrieval models place such pairs closer because they match on topic regardless of stance. For CRW, where a question and its negation measure the same policy

dimension, the retrieval models' partial negation invariance is an advantage.

III. PROBLEM SETUP & DEFENSE MECHANISM

This section introduces the tools required to formalize redundancy attacks in VAAs, as well as the proposed defense mechanism.

a) VAA Model: We model a VAA as a mapping from policy positions of both voters and candidates to personalized rankings over candidates for each voter. Formally, let $\mathcal{Q} = \{q_t\}_{t=1}^Q$ denote the set of survey questions, $\mathcal{V} = \{v_i\}_{i=1}^V$ the set of voters, and $\mathcal{C} = \{c_j\}_{j=1}^C$ the set of candidates. Each question q_t is associated with a finite set of admissible answers $A_t \subseteq [0, 100]$. We then represent each individual $x_i \in \mathcal{V} \cup \mathcal{C}$ with its response profile $\mathbf{x}_i = [x_i(q_1), \dots, x_i(q_Q)]^\top \in \prod_{t=1}^Q A_t$, where $x_{i,t} := x_i(q_t) \in A_t$ denotes the response of individual x_i to question q_t . Additionally, each voter $v_i \in \mathcal{V}$ may assign personalized importance weights to survey questions in the form of a probability distribution $w_i \in \Delta(\mathcal{Q})$, where $w_{i,t} \in [0, 1]$ denotes the weight assigned to question q_t and $\sum_t w_{i,t} = 1$. When personalized weights are not provided, the system defaults to uniform prior weights $w_{i,t} = w_t^{\text{unif}} := 1/Q$, for each question $q_t \in \mathcal{Q}$. Recommendation is then based on a weighted ℓ_p distance, with $p \geq 1$: for each voter v_i , the VAA evaluates distances $d_p(\mathbf{v}_i, \mathbf{c}_j, \mathbf{w}_i) = (\sum_{t=1}^Q w_{i,t} |v_{i,t} - c_{j,t}|^p)^{1/p}$ between v_i and each candidate $c_j \in \mathcal{C}$. Candidates are then ranked in increasing order of distance, yielding a ranking r_i over $\mathcal{C}$. We interchangeably view r_i as a total order over candidates and as a function $r_i : \mathcal{C} \rightarrow \{1, \dots, |\mathcal{C}|\}$ assigning positions consistent with this ordering. Note that questions can equivalently be viewed as a pair of response vectors over voters and candidates, namely $q_t := (q_t^V, q_t^C)$, where $q_t^V = [v_1(q_t), \dots, v_V(q_t)]^\top$ and $q_t^C = [c_1(q_t), \dots, c_C(q_t)]^\top$. Assuming uniform prior weights w^{unif} for all voters, the resulting ranking $r_i(\mathcal{Q})$ is then fully determined by the questionnaire $\mathcal{Q}$.

b) Approximate Question Clones: Considering questions in vector form $q_t = (q_t^V, q_t^C)$, one may equip the ambient space (whether $\mathbb{R}^V$, $\mathbb{R}^C$, or $\mathbb{R}^{V+C}$) with a standard distance function $d : \mathbb{R}^{V+C} \times \mathbb{R}^{V+C} \mapsto \mathbb{R}_{\geq 0}$ satisfying identity of indiscernibles, i.e., $d(x, y) = 0 \iff x = y$, symmetry, i.e., $d(x, y) = d(y, x)$ and triangle inequality, i.e., $d(x, z) \leq d(x, y) + d(y, z)$ for all $x, y, z \in \mathbb{R}^{V+C}$. Note however that the space of questions exhibits additional structure induced by negation: if users were fully consistent, the response vector to “Taxes should be increased” could be perfectly inferred from that of “Taxes should be decreased”. We hence define the $[0, 100]$ -involution $\sigma : x \mapsto 100 - x$ extended component-wise, and restrict our attention to pseudo-metrics¹ d^{beh} that remain stable on the orbits of σ , i.e., $d^{\text{beh}}(x, y) = d^{\text{beh}}(x, \sigma(y))$.² For example, $d_p^{\text{beh}}(x, y) = \min\{d_p(x, y), d_p(x, \sigma(y))\}$ defines such a pseudo-metric, where d_p denotes the standard ℓ_p

distance for $p \geq 1$. For a threshold value $\alpha \geq 0$, we then say that questions q and q' in $\mathcal{Q}$ are α -behavioral clones when $d^{\text{beh}}(q, q') \leq \alpha$.

However, this view on questions admits limitations: first, even questions that are arguably truly identical, e.g., strict negations, may elicit different responses in practice, e.g., due to the *acquiescence bias* [9]. Second, two questions that are close according to d^{beh} may still concern fundamentally different policy topics, with proximity driven by shared response patterns among a majority of respondents (e.g., carbon taxation and public transport investment). For this reason, one may want to generalize the concept of clones and consider embedding questions in a representation space $\mathcal{E}$ via an embedding function $E : \mathcal{Q} \mapsto \mathcal{E}$, e.g., based only on their raw text. We then equip $\mathcal{E}$ with a pseudo-metric $d^{\mathcal{E}} : \mathcal{E} \times \mathcal{E} \mapsto \mathbb{R}_{\geq 0}$ yielding a pseudo-metric space that can be interpreted either as an idealized policy space or as a practical instantiation via sentence embeddings induced by a large language model. For a fixed embedding function E and threshold $\alpha \geq 0$, we then similarly define α -embedding clones as questions q and q' in $\mathcal{Q}$ such that $d^{\mathcal{E}}(E(q), E(q')) \leq \alpha$.

c) Redundancy Attacks: As noted in previous work [2], modifying a questionnaire can significantly affect recommendations. Consider a question $q \in \mathcal{Q}$, let $q'_1, \dots, q'_\ell$ be ℓ distinct 0-behavioral clones of q , and define the manipulated questionnaire $\mathcal{Q}' = \mathcal{Q} \cup \{q'_1, \dots, q'_\ell\}$. For voters without personalized weights, the system defaults to uniform w^{unif} and the effective weight assigned to q increases from $\frac{1}{Q}$ under $\mathcal{Q}$ to $\frac{1+\ell}{Q+\ell}$ under $\mathcal{Q}'$. When $Q \gg \ell$, the influence of q on distance computations, and hence recommendations, is multiplied by a factor of roughly ℓ . While exact (0-behavioral) clones may be unrealistic, the same effect persists for α -behavioral clones with $\alpha \ll 1$. As a result, even a well-intentioned questionnaire designer may inadvertently distort recommendations by introducing questions that induce similar response patterns. A strategic question designer, however, may exploit this mechanism to deliberately increase the visibility of a targeted party by injecting approximate clones of favorable questions.

Formally, an adversary transforms the original questionnaire $\mathcal{Q}$ into $\mathcal{Q}' = \mathcal{Q} \cup \mathcal{Q}_{\text{clone}}$, where $\mathcal{Q}_{\text{clone}}$ consists of α -behavioral³ clones of target questions in $\mathcal{Q}$. The scale of the attack is hence characterized by the number of targeted questions n , the number of clones per targeted question ℓ (resulting in $|\mathcal{Q}_{\text{clone}}| = n\ell$), and the clone-threshold α . As a result, the VAA computes rankings⁴ $r_i(\mathcal{Q}')$ instead of the original $r_i(\mathcal{Q})$.

The severity of the attack can then be quantified in different manners. At the individual level, one may measure the overlap between the top- k candidates in voter v_i 's ranking before and

¹Formally, a pseudo-metric exchanges the *identity of indiscernibles* condition for the weaker *identity*, i.e., $d(x, x) = 0$.

²We assume that the admissible response sets A_t are stable under σ .

³It is unclear how embedding clones that are not behavioral ones would affect recommendations.

⁴Assuming all voters use uniform prior weights w^{unif} .

after perturbation using the *Jaccard similarity*

$$J_k(r_i(\mathcal{Q}), r_i(\mathcal{Q}')) := \frac{|T_k(r_i(\mathcal{Q})) \cap T_k(r_i(\mathcal{Q}'))|}{|T_k(r_i(\mathcal{Q})) \cup T_k(r_i(\mathcal{Q}'))|},$$

where $T_k(r_i)$ denotes the set of the top- k candidates in ranking r_i . Alternatively, the global consistency between v_i 's rankings can be assessed using *Spearman's rank correlation*, i.e.,

$$\rho_S(r_i(\mathcal{Q}), r_i(\mathcal{Q}')) := 1 - \frac{6 \sum_{j=1}^C (r_i(\mathcal{Q})(c_j) - r_i(\mathcal{Q}')(c_j))^2}{C(C^2 - 1)},$$

which captures agreement over the full candidate list of size C . At the population level though, what matters is the electoral impact for each party. We adopt the k -visibility metric from [2], defined as the average fraction of top- k recommendation slots occupied by candidates from party p , i.e.,

$$\nu_k(p, \mathcal{Q}) := \frac{1}{V_k} \sum_{v_i \in \mathcal{V}} |T_k(r_i(\mathcal{Q})) \cap \mathcal{C}_p|,$$

where $\mathcal{C}_p \subseteq \mathcal{C}$ denotes the set of candidates belonging to party p . We then define the k -visibility distortion of party p $\Delta\nu_k(p) = \nu_k(p, \mathcal{Q}') - \nu_k(p, \mathcal{Q})$ as the shift in party visibility between the original and the tempered questionnaires.

d) CRW Defense Mechanism.: The clone-robust weighting framework [3, 4] addresses redundancy in metric spaces by enforcing three principles on the prior weight distribution: symmetry, continuity, and locality—ensuring that similar items only shift weights locally. In our context, this guarantees that a group of near-duplicate questions collectively receives the same total weight as a single isolated question. We adopt the graph-based construction from [4]. Formally, consider the set of embedded questions $S = \{E(q_t)\}_{t=1}^Q$ in pseudo-metric space $(\mathcal{E}, d^{\mathcal{E}})$. A *clone-robust weighting function* $f_{\nu, w}$ assigns each question a weight by

$$f_{\nu, w}(S)(q_t) := \int_{\mathbb{R}_{\geq 0}} \nu(r) w(G_r(S))(q_t) dr,$$

where ν is a probability density function with support on an interval $[0, \alpha]$, the r -neighborhood graph $G_r(S)$ denotes the undirected graph with vertex set S and edges between all pairs of questions within distance r , and w is a *clone-robust graph weighting function* ensuring weight-sharing across neighbors only, c.f. [4, Definition 2]. Intuitively, clone-robust weights are first computed for each neighborhood graph $G_r(S)$ and then aggregated over radii according to distribution ν .

In this work, we take the most simple instantiation of clone-robust weighting function and use a uniform prior $\nu_{\alpha}(r) = 1/\alpha$ over the radii in $[0, \alpha]$, as well as the class uniform graph weighting function w^{CU} from [4]. This clone-robust graph weighting function relies on the equivalence relation $\equiv_G$ induced by the closed neighborhoods of a graph, i.e., two nodes $u, v \in V(G)$ are equivalent if $N_G(u) \cup \{u\} = N_G(v) \cup \{v\}$. Weights are computed by first assigning equal mass to each equivalence class, and then distributing it uniformly within each class, i.e.,

$$w^{\text{CU}}(G)(u) := \frac{1}{|V/\equiv_G| \cdot |[u]_G|},$$

where $[u]_G$ denotes the equivalence class of node $u \in V(G)$ with respect to $\equiv_G$ and $V/\equiv_G = \{[u]_G\}_{u \in V(G)}$ denotes the set of equivalence classes.

For each question $q_t \in \mathcal{Q}$, this hence yields clone-robust prior weights $w_t^{\text{CRW}}(\mathcal{Q}) = f_{\nu_{\alpha}, w^{\text{CU}}}(E(\mathcal{Q}))(q_t)$, which can be evaluated in time complexity $O(Q^4)$, c.f. [4]. When used to compute distances $d_p(\mathbf{v}_i, \mathbf{c}_j, w_t^{\text{CRW}}(\mathcal{Q}))$, these weights then induce rankings $r_i^{\text{CRW}}(\mathcal{Q})$. The impact of an attack $\mathcal{Q}'$ is then measured by comparing $r_i^{\text{CRW}}(\mathcal{Q})$ and $r_i^{\text{CRW}}(\mathcal{Q}')$, for instance with the Jaccard similarity J_k or Spearman's rank correlation ρ_S . At the party level, we define the corresponding k -visibility as $\nu_k^{\text{CRW}}(p, \mathcal{Q}) := \frac{1}{V_k} \sum_{v_i \in \mathcal{V}} |T_k(r_i^{\text{CRW}}(\mathcal{Q})) \cap \mathcal{C}_p|$, and the associated visibility distortion $\Delta\nu_k^{\text{CRW}}(p) := \nu_k^{\text{CRW}}(p, \mathcal{Q}') - \nu_k^{\text{CRW}}(p, \mathcal{Q})$.

While the defense mechanism accommodates diverse design choices, any concrete instantiation requires specifying three components: (i) the embedding function E , (ii) the distance in embedding space $d^{\mathcal{E}}$, and (iii) the integration radius α used in CRW. The choice of E and $d^{\mathcal{E}}$ is clearly critical, as it determines which questions are deemed redundant and thus which redundancies CRW corrects. The role of α , though seemingly secondary, is equally important: it defines the interaction radius so that only α -embedding clones can share weight. If α is too small, correction is limited to the closest questions, reverting to uniform weights in the absence of exact clones; if too large, all questions collapse into a single equivalence class, again recovering uniform weighting.

IV. DATASET

SmartVote is Switzerland's primary Voting Advice Application, operated by the non-profit association Politools [19]. In the 2023 Swiss National Council elections, it was used by up to 20% of eligible voters and 85% of all registered candidates [19], making it a particularly relevant testbed for studying adversarial robustness in VAAs. We hence empirically validate the proposed defense pipeline on the SmartVote dataset for the 2023 Swiss National Council elections in Canton Zurich, collected and anonymized by Politools [19] in accordance with applicable Swiss data protection laws. This dataset comprises responses and metadata from $V = 41,658$ voters and $C = 1,029$ candidates over $Q = 75$ questions, with admissible response sets $A_t = \{0, 25, 75, 100\}$ for policy questions ($1 \leq t \leq 60$), $A_t = \{0, 17, 33, 50, 67, 83, 100\}$ for value questions ($61 \leq t \leq 67$), and $A_t = \{0, 25, 50, 75, 100\}$ for budget questions ($68 \leq t \leq 75$). Recommendations are based on a weighted ℓ_2 metric $d_{\text{SmartVote}}(\mathbf{v}_i, \mathbf{c}_j, \tilde{\mathbf{w}}_i) = (\sum_{t=1}^Q (\tilde{w}_{i,t}(v_{i,t} - c_{j,t}))^2)^{1/2}$. Note however that this differs from the formulation in Section III in two ways. First Smartvote weights $\tilde{w}_{i,t}$ are unnormalized and selected by the users from the coarse set $W_t = \{0, 0.5, 1, 2\}$ (with default value 1 for all questions). Second, the contribution of weights $\tilde{w}_{i,t}$ for each question is quadratic, whereas that of $w_{i,t}$ in Section III is linear. Since normalization does not affect rankings, the two formulations are equivalent up to the bijection $x \mapsto x^2$. For each voter, candidates are then ranked by

increasing distance, and the top 36 candidates—corresponding to the number of National Council seats allocated to Canton Zurich—are prominently displayed. Accordingly, we evaluate distortion using top- k metrics with $k = 36$, namely Jaccard similarity J_{36} and party visibility distortion $\Delta\nu_{36}$.

V. EVALUATION OF A SEMANTIC-BASED DEFENSE

In this section, we instantiate and evaluate a semantic-based defense pipeline. We consider several widely used sentence embedding models for the embedding function $E : \mathcal{Q} \mapsto \mathcal{E}$ and measure distances in the embedding space using the normalized Euclidean distance $d^{\mathcal{E}}(q, q') = \|\hat{E}(q) - \hat{E}(q')\|_2$, where $\hat{E}(q) = E(q)/\|E(q)\|_2$ denotes the normalized embedding of q . This choice is motivated by the fact that these models are usually trained with objectives based on normalized cosine similarity, which is monotonically related to $d^{\mathcal{E}}$ via $d_{\text{emb}}(q, q') = \sqrt{2(1 - \cos(\theta_{q, q'}))}$, where $\theta_{q, q'}$ denotes the angle between the embeddings of q and q' . The choice of α is finally treated as a tunable hyper-parameter.

A. Methodology

We describe hereafter the empirical strategy used to instantiate and evaluate the semantic-based defense mechanism. This evaluation proceeds in six stages.

a) Vulnerability Assessment: We quantify the baseline vulnerability by measuring the distortion in rankings under the addition of 0-embedding 0-behavioral clones, and identify the most sensitive questions. For each of the 75 questions, we craft a redundancy attack where $\mathcal{Q}_{\text{clone}}$ contains ℓ exact copies of the targeted question (0-embedding and 0-behavioral clones), for increasing ℓ values in $[1, 30]$. We then measure the resulting ranking distortion (J_{36}, ρ_S) under uniform weights.

b) Synthetic Clones Generation: To model more sophisticated attacks, we generate synthetic embedding (but 0-behavioral) clones of varying types and difficulties for each question in the dataset. Synthetic clones are generated by prompting GPT-4o [18] to produce 4 types of transformations: *Easy paraphrase*: minor rewording with significant lexical overlap, preserving the core structure. These serve as a baseline for detecting near-verbatim redundancy. *Hard paraphrase*: fully rephrased with no shared content words, conveying the identical policy position. This tests whether embeddings capture conceptual meaning beyond lexical overlap. *Easy negation*: a direct negation of the easy paraphrase, designed to assess whether opposing stances are still mapped to the same policy dimension. *Hard negation*: a negated version of the hard paraphrase, representing the most challenging case for text-based similarity. For each question, we generate 4 instances per transformation type (16 in total). Importantly, all synthetic clones are assumed to be 0-behavioral, i.e., they inherit the response pattern of the original question, with negated versions adjusted via the transformation σ from Section III. Full prompting details, along with an example of all four clone types for a representative question, are provided in the full version. Based on these variants, we define 5 attack settings, each injecting 4 clones per question: *Full easy*

paraphrase, *Full hard paraphrase*, *Full easy negation*, *Full hard negation*, and a *Mixed* setting. The “Full” attack settings inject all four clones of the corresponding type for all targeted questions, while the mixed setting includes one variant of each type.

c) Model and α Selection: We preselected a set of ten sentence embedding models spanning a broad range of training paradigms and architectures, from symmetric semantic similarity models (e.g., SBERT [21]) to asymmetric retrieval models (e.g., E5-Asym), as well as versatile hybrid models (e.g., E5 [27], BGE-M3 [5], GTE [13]). In addition, the selection includes models that support task-specific instructions—namely E5-Instruct [28], E5-Asym-Instruct [28], Jina-v3 [22], Qwen3-Embedding [29], and Nomic-v2 [17]. For these models, we prepend a short directive framing the task as clustering political survey questions by topic, while the remaining models operate directly on raw question text. To craft redundancy attacks, we restrict attention to the five most impactful questions identified in Step 1, as these yield the strongest distortions and thus provide a higher signal-to-noise ratio for model comparison. We then generate a total of 5 attack sets $\mathcal{Q}_{\text{clone}}$ by adding 4 clones for each of the 5 targeted questions according to the attack settings defined in Step 2, and define the corresponding manipulated questionnaire as $\mathcal{Q}' = \mathcal{Q} \cup \mathcal{Q}_{\text{clone}}$. For every model-attack pair, we sweep the CRW radius α over 21 values in the interval $[0.01, 3.0]$ and aggregate each robustness metric across voters and attack conditions. We then select the optimal α^* for each model by maximizing a composite score over these metrics. We finally rank the pairs (model, α^*) on each metric and select the model with the lowest mean rank across metrics. See the full version for more details.

d) Robustness on the Full Questionnaire: For the pair (model, α^*) selected in Step 3, we conduct a full questionnaire robustness evaluation to assess both generalization and hyperparameter stability. We construct five distinct attack sets $\mathcal{Q}_{\text{clone}}$ for each of the 75 survey questions based on the five attack settings in Step 2 (four clones inserted in each), yielding a total of 375 manipulated questionnaires $\mathcal{Q}'$. We then repeat the α sweep over 21 values in $[0.01, 3.0]$ for each $\mathcal{Q}'$ and benchmark CRW’s ranking robustness against the uniform weight baseline. This procedure ensures that the selected α^* reflects a stable geometric property of the embedding space rather than an artifact of the targeted question subset in Step 3, and validates that the defense mechanism generalizes across the full questionnaire.

e) Robustness against Targeted Party Manipulation: We next quantify the real-world electoral significance of both the attack and its mitigation and consider redundancy attacks deliberately designed to favor a given party’s visibility. We focus on the six major parties—the *Social Democratic Party* (SP), the *Green Party* (Green), the *Green Liberal Party* (GLP), *The Centre* (Centre), *FDP. The Liberals* (FDP), and the *Swiss People’s Party* (SVP)—which collectively span the Swiss political spectrum and account for the vast majority of SmartVote usage. For each party p , we identify the 5 target questions whose cloning most increases that party’s visibility $\nu_{36}(p, \mathcal{Q}')$. These targets

are selected by simulating for each question $q \in \mathcal{Q}$ an idealized attack where $\mathcal{Q}'$ is obtained by injecting four exact copies—0-embedding and 0-behavioral clones—of q . We then generate two adversarial attacks per party: (i) a *minimal attack*, where we inject one easy paraphrase per target question ($|\mathcal{Q}_{\text{clone}}| = 5$) to simulate a low-resource adversary using the most natural and detectable clone type; (ii) a *maximal attack*, where we inject one clone of each type for every target question (mixed setting, $|\mathcal{Q}_{\text{clone}}| = 20$) to represent a well-resourced adversary employing all four cloning strategies simultaneously. For each party and attack setting, we compare the visibility distortion under CRW, $\Delta\nu_{36}^{\text{CRW}}$, with the uniform-weight baseline, $\Delta\nu_{36}$.

f) *Generalization to Approximate Behavioral Clones*: Finally, we revisit Steps 4 under a more realistic setting where clones are only approximately behavioral. For each of the 75 questions, we construct a mixed attack set (four clones, one per type), and perturb responses using a global noise parameter λ . Concretely, for each clone q' of a question q_t , each entry $x(q')$ is independently modified with probability λ across all users $x \in \mathcal{V} \cup \mathcal{C}$. When modified, the new response is sampled from A_t with probability inversely proportional to its distance from the original value, i.e., $P(a_{\text{new}} | a_{\text{old}}) \propto |a_{\text{new}} - a_{\text{old}}|^{-1}$. We evaluate robustness by sweeping over 8 values of $\lambda \in [0, 1]$, repeating each experiment 20 times to ensure statistical significance.

B. Results

All experiments are conducted on an Intel Xeon E5-2690 processor, using 4 cores and 32GB of RAM. Each data point (two recommendation runs plus Jaccard and Spearman computations across all voters) takes about two minutes; CRW computations are negligible.

a) *Vulnerability Assessment*: Figure 1a demonstrates that recommendation distortion scales monotonically with adversarial effort across every question in the survey. Tracking the maximum (Q32214), median (Q32242), and minimum (Q32235) impact questions (at $\ell = 4$ clones) reveals that Top-36 Jaccard similarity degrades immediately upon the introduction of even a single clone. While the magnitude varies—reducing Jaccard similarity to 0.657 for the most vulnerable question and 0.928 for the most robust—the universal decay confirms that cloning is a structural vulnerability of the questionnaire rather than an artifact of specific polarizing topics. Because Jaccard similarity and Spearman’s correlation are highly correlated ($\rho = 0.965$, see full version), we report only the former for the remainder of this paper.

b) *Model and α Selection*: Without CRW mitigation, injecting four clones for each of the five most impactful questions reduces mean Jaccard similarity to 0.452, displacing nearly half of each voter’s top-36 recommendations. As shown in Figure 1b, CRW improves robustness across all ten models, though their response to α varies structurally. Top-performing models such as E5-Instruct and Jina-v3 exhibit sharply peaked profiles with narrow 95%-of-maximum α intervals lengths (0.23 and 0.35, respectively), while lower-ranked models like SBERT and E5-Asym-Instruct display much flatter pro-

files, maintaining near-optimal performance across effective α ranges of width 2.28 and 2.99.

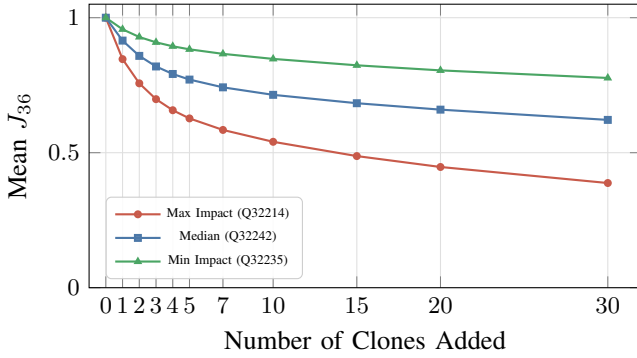
The final model hierarchy, detailed in Table I, identifies E5-Instruct as the clear winner. It achieves the top rank across both J_{36} and ρ_S and is followed closely by Jina v3, while Qwen3 and GTE occupy the middle tier. The performance gap is most pronounced in the Jaccard metric, where values range from E5-Instruct’s 0.585 down to 0.491 for E5-Asym-Instruct. While this margin narrows for Spearman’s ρ_S global metric—spanning 0.952 to 0.921—every model at its optimal α^* provides a substantial recovery over the unweighted baseline, confirming that the defense consistently improves ranking stability across the full candidate list.

Model	$J_{36}^{\text{CRW}} \uparrow$	$\rho_S^{\text{CRW}} \uparrow$	Mean rank
E5-Instruct	0.585	0.952	1
Jina v3	0.581	0.951	2
Qwen3	0.565	0.946	3
GTE	0.563	0.945	4
BGE-M3	0.552	0.942	5
Nomic v2	0.542	0.939	6.5
SBERT	0.541	0.940	6.5
E5	0.530	0.936	8
E5-Asym	0.518	0.932	9
E5-Asym-Instruct	0.491	0.921	10
Baseline	0.452	0.905	11

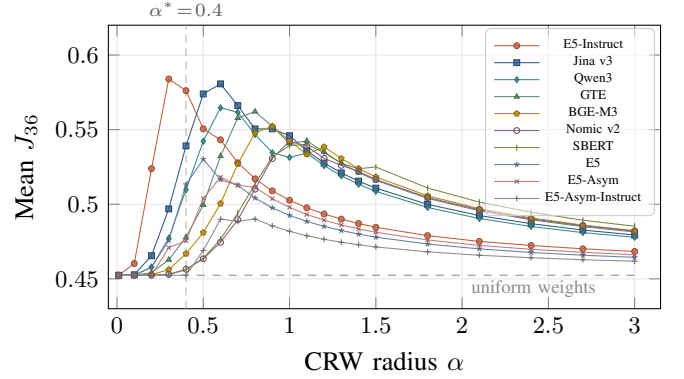
Table I: Embedding model performance at optimal α^* , ordered by mean rank across J_{36} and ρ_S , averaged over all voters and five clone conditions.

c) *Robustness on the Full Questionnaire*: After identifying an optimal aggregate radius $\alpha^* = 0.4$ during model selection, we find that this threshold transfers stably to the full 75-question survey (see full version), suggesting it reflects a structural property of the E5-Instruct embedding space rather than an artifact of the target set. Table II shows that CRW improves robustness across all attack types, with mean Jaccard gains decreasing monotonically with attack difficulty: from +0.089 for easy paraphrases to +0.032 for hard negations, reflecting the reduced separability of more sophisticated attacks in embedding space. In relative terms, the defense absorbs up to 47% of the distortion for the most detectable attacks, and still recovers 17% to 25% under the hardest conditions. Note that this clear hierarchy in attack settings is consistent with the clone generation procedure: paraphrases are more effectively mitigated than negations, and hard variants systematically outperform easy ones at evading the defense.

Figure 2 refines this aggregate view by showing the per-question CRW improvement ΔJ_{36} at $\alpha^* = 0.4$ across all 375 question–attack combinations. Consistent with the overall results, 370 of the 375 cases exhibit strictly positive improvement, with the remaining 5 being negligible (maximum degradation < 0.001) and confined to the hardest attack settings, confirming that the mitigation generalizes across the questionnaire. At the same time, the distribution of improvements varies systematically with attack difficulty: easy paraphrases exhibit the widest spread (up to +0.156), while



(a) Recommendation distortion scales with adversarial effort. The top-36 Jaccard similarity, averaged across all voters, decays predictably as identical clones are inserted. Spearman’s ρ evaluated over the full candidate list ($n = 1,029$) exhibits a similar decay pattern.



(b) Mean Jaccard similarity as a function of CRW radius α for all 10 embedding models, averaged over all five clone conditions. The dashed horizontal line marks the uniform-weight baseline ($J = 0.452$). Models are listed in the legend by composite rank.

Figure 1: Experimental Results for Steps 1 and 3 in Section V-A.

Attack Setting	$J_{36}^{\text{CRW}} \uparrow$	$\Delta J_{36} \uparrow$	Mitigated (%)
Easy paraphrase	0.900	+0.089	47.1%
Easy Negation	0.892	+0.081	42.8%
Mixed	0.857	+0.047	24.9%
Hard paraphrase	0.857	+0.046	24.3%
Hard Negation	0.843	+0.032	16.9%

Table II: Mean CRW robustness J_{36}^{CRW} and improvement $\Delta J_{36} = J_{36}^{\text{CRW}} - J_{36}^{\text{unif}}$ by attack setting across all 75 questions, using E5-Instruct at $\alpha^* = 0.4$. Baseline $J_{36}^{\text{unif}} = 0.811$ across all conditions.

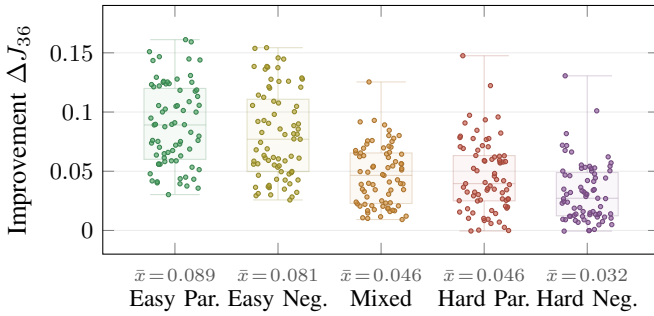


Figure 2: Per-question CRW improvement in Jaccard similarity (top-36) across all 75 questions and 5 attack types (375 points), using E5-Instruct at $\alpha^* = 0.4$. Each dot represents one question, and overlaid box plots summarize the distribution; mean values are reported in the legend.

both magnitude and variability decrease under harder conditions. This suggests that degradation becomes more uniform across questions rather than concentrated on specific failures, although a few outliers still achieve substantial gains even in the hardest settings.

d) Robustness against Targeted Party Manipulation: Targeted cloning induces systematic and non-trivial distortions in party visibility even at low attack budgets. As shown in Table III, the

defense mechanism consistently reduces these artificial gains across all parties and attack scenarios. Under the minimal attack setting, mitigation ranges from 42.6% (Centre) to 61.8% (GLP), with a mean reduction of 55.8%. Under the maximal attack setting, the defense provides partial but consistent mitigation, ranging from 19.4% (Green) to 31.7% (SVP), with a mean of 26.7%. Importantly, the correction operates jointly across the system: reductions in the targeted party’s inflated visibility are accompanied by proportional recovery for adversely affected parties. For instance under the maximal attack favoring the Centre, CRW reduces Centre’s artificial gain from +4.76pp to +3.49pp, while proportionally restoring SP’s loss (from −2.35pp to −1.76pp) and Green’s loss (from −2.39pp to −1.64pp). The weaker performance in the maximal attack setting stems from the attack’s heterogeneous design, which combines multiple clone types, including the most challenging variants. These harder clones are naturally dispersed further from their originals in the embedding space, reducing the model’s ability to detect redundancy.

Party	Maximal (4× mixed)			Minimal (1× easy)		
	$\Delta \nu_{36}$	$\Delta \nu_{36}^{\text{CRW}}$	Mitig.	$\Delta \nu_{36}$	$\Delta \nu_{36}^{\text{CRW}}$	Mitig.
Centre	+4.76	+3.49	26.7%	+1.83	+1.05	42.6%
SP	+4.53	+3.37	25.7%	+1.76	+0.83	53.0%
GLP	+4.16	+2.92	29.8%	+1.74	+0.67	61.8%
Green	+2.20	+1.77	19.4%	+0.99	+0.41	58.8%
FDP	+1.44	+1.05	26.9%	+0.58	+0.24	58.1%
SVP	+1.37	+0.93	31.7%	+0.47	+0.19	60.5%
Mean	+3.08	+2.26	26.7%	+1.23	+0.56	55.8%

Table III: Party visibility changes under targeted cloning attacks and CRW correction. $\Delta \nu_{36}$ and $\Delta \nu_{36}^{\text{CRW}}$ report changes relative to the unmanipulated baseline under uniform weights and CRW (E5-Instruct, $\alpha^* = 0.4$). “Mitig.” denotes the fraction of induced distortion removed by CRW.

e) Generalization to Approximate Behavioral Clones: As depicted in Figure 3, the proposed defense generalizes grace-

fully to a setting with approximate behavioral clones. Indeed, increasing the noise level λ predictably degrades recommendation robustness as injected clones drift away from their behavioral target— J_{36}^{CRW} dropping from 0.82 at zero noise $\lambda = 0$ to 0.73 when every entry is perturbed, i.e., $\lambda = 1$. Crucially though, CRW consistently improves robustness across all noise levels and keeps a remarkably stable advantage over the baseline, averaging 0.07 improvement across the full range of λ . This suggests that the CRW defense does not merely rely on the perfect structure of exact behavioral clones, and offers protection even under realistic levels of behavioral noise. It is also worth noting that even at $\lambda = 1$, performance does not collapse entirely. This is a direct consequence of the perturbation model: larger deviations are assigned lower probability, so even fully perturbed clones retain partial behavioral similarity, limiting the overall degradation.

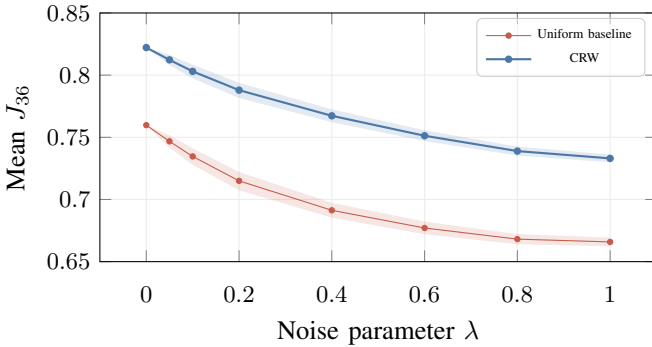


Figure 3: Robustness to approximate behavioral clones under the mixed attack setting, where each response entry is independently modified with probability λ . Each point averages the top-36 Jaccard similarity over all voters, 75 questions and 20 repetitions; shaded bands indicate the 90% confidence interval.

VI. DISCUSSION

In this section, we discuss our findings, their interpretation, and the role of a defense mechanism.

a) Main findings: Our results confirm that the vulnerability identified by Berdoz et al. [2] is indeed potent: injecting even a single clone suffices to measurably distort voter-candidate recommendations, with effects that scale monotonically as additional clones are introduced. The semantic-based defense instantiated in Section V using sentence embedding models systematically improves robustness to such redundancy attacks across all embedding models, radii α , clone types, questions and behavioral noise, mitigating 17–47% of the induced distortion. At the party level, CRW improves robustness against targeted attacks designed to favor specific parties, reducing artificial party visibility gains by 43–62% under a minimal one-clone attack and by 19–32% under worst-case multi-clone attacks.

b) Meaning of Redundancy in a VAA: While the defense pipeline in Section V shows promising results, it only addresses a narrow, text-based notion of redundancy. First, semantic embeddings are inherently vulnerable to evasion:

sufficiently complex clones—double negations, convoluted paraphrases, or jailbreak-style rewrites—can always bypass them. Second, text similarity may miss deeper logical dependencies: expanding the army may be tightly coupled to increasing the defense budget, and positions on environmental issues may largely reduce to prior beliefs about anthropogenic climate change. Both limitations suggest response patterns as a more robust basis for redundancy. Third, the current approach deliberately avoids population-level responses so that majority views do not dominate minority ones. Whether redundancy should depend on cultural context is nonetheless a normative question. We argue that any universal notion of political proximity is likely untenable: Switzerland (land-locked) and Indonesia (archipelagic) may reasonably disagree on whether “Boat infrastructures” and “Transportation of goods” are closely related topics. Setting this issue aside, replacing uniform weights with a behaviorally informed prior naturally leads to a metric-learning perspective. One may then seek a pseudo-metric $\hat{d}^{\text{beh}}$ minimizing the expected prediction error up to negation

$$\mathcal{L}_{s,t}(\hat{d}^{\text{beh}}) = \min_{f \in \{\sigma, \text{Id}\}} \mathbb{E}_{x_i \sim \mathcal{V} \cup \mathcal{C}} [d_i(q_s, f(q_t)) - \hat{d}^{\text{beh}}(q_s, f(q_t))]^2$$

where $d_i(q_s, f(q_t)) = |x_{i,s} - f(x_{i,t})|$ is the distance between q_s and q_t according to x_i . In this view, $d_1^{\text{beh}}(q, q') = \min\{d_1(q, q'), d_1(q, \sigma(q'))\}$ arises naturally, as it corresponds (up to scaling) to the empirical average distance between answers to q and q' across voters, up to negation. The choice of α is then fundamentally normative; for instance, it could reflect the fraction of the population that must fully agree in response patterns between two questions to ensure that they start sharing weight.

Such a data-driven approach gives rise to a classical cold-start problem: weights are required to generate recommendations for voters, yet voters responses are needed in the first place to compute distances and thus determine these weights. In practice, this issue can be mitigated by first computing distances using candidate responses, and then updating them in an online fashion as voter data becomes available. To simulate this setting, we randomly split the voter dataset and estimate distances using all candidate responses but only 5% of voter responses. Figure 4 reports the resulting distortion in recommendations for the remaining voters, relative to the uniform-weight baseline and across a range of possible choices for α . While question weights exhibit some variation (with up to a 25% decrease for the smallest weight), the changes in recommendations it induces remain limited. This reflects the careful and well-balanced design of the SmartVote questionnaire, with little inherent redundancy, and suggests that the instantiated behavioral-based defense operates conservatively in such settings.

c) Role of the CRW Weights: In this paper, we argue that CRW should serve as a *default alternative to uniform weighting*, to be *replaced by voter-specific weights when available*. A natural alternative would be to apply CRW on top of personalized weights (e.g., multiplicatively with SmartVote weights $\tilde{w}_{i,t}$).

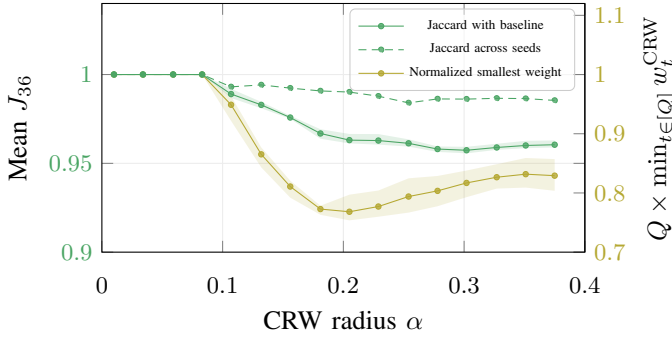


Figure 4: Change in recommendations under CRW with behavioral distance d_1^{beh} . For each seed, d_1^{beh} is computed using all candidate responses and a random 5% subsample of voters. Green lines (left axis) report Jaccard similarities (J_{36}) across radii: the solid line compares recommendations under uniform vs. CRW weights, while the dashed line shows agreement across voter splits; both are averaged over the remaining 95% of voters (and across seed pairs for the latter). The yellow line (right axis) shows the minimum question weight, normalized so weights sum to Q . Shaded bands indicate the range across all five seeds. Distances are normalized so that $\alpha = 1$ corresponds to the 75th percentile of pairwise question distances.

The intuition is that users assign weights to questions individually, based on perceived importance of each topic, without accounting for the global normalization effect induced by the full weight vector. As a result, duplicated questions may each be assigned high importance independently, even though this effectively dilutes the relative weight of all other questions after normalization. Such a multiplicative scheme would in principle preserve robustness to cloning by cancelling out duplicated questions—unlike the formulation considered here—but at the cost of reduced transparency in the recommendation process. Moreover, “errors” in semantic distance estimation would become effectively irreversible, making the choice of metric more critical. In practice, subtle but meaningful differences between questions may justify distinct weights from informed users.

d) Improving Personal Weights Elicitation: The defense pipeline we propose compensates for the absence of personalized question weights for most users, addressing symptoms rather than root causes. The underlying issue lies with VAA platforms, which must better elicit voter preferences despite user fatigue. We highlight three directions: (i) introduce mandatory weight elicitation as a dedicated step (as in Wahl-O-Mat); (ii) collect coarse topic-level preferences upfront—when engagement is highest—followed by post-completion refinement at both topic and question levels; and (iii) represent weights as normalized distributions to improve interpretability, making trade-offs explicit. This could be supported by interactive visualizations (e.g., dynamic spider-web interfaces), with CRW-derived weights serving only as the default.

e) Limitations & Future Work: As all our experiments rely on a single dataset—SmartVote 2023 Canton Zurich—evaluating CRW on other VAAs (e.g., Wahl-O-Mat or Vote Compass) would test transferability across electoral systems, languages, and questionnaire designs. Such variation is likely to affect both optimal embedding models and CRW hyperparameters; a systematic study is left for future work. A second direction is fine-tuning embedding models for this setting, either by enforcing consistency across multilingual versions of the same questionnaire or via domain-specific adaptation of political structure. Lee et al. [12] show that fine-tuning SBERT on political data captures belief correlations, suggesting a natural route to better align semantic and behavioral distances. Finally, several CRW design choices remain open. Alternative radius distributions $\nu(r)$ could concentrate weight on empirically relevant scales, while more expressive graph constructions, such as maximal clique-based methods [4], may better handle dense questionnaires. The original Euclidean formulation [3], based on smooth volume overlap rather than binary neighborhoods, offers a complementary alternative. Understanding how these variants interact with embedding choices remains open.

VII. CONCLUSION

As VAAs increasingly influence democratic outcomes, their algorithmic integrity is a matter of critical public interest. Structural vulnerabilities in these platforms can enable subtle yet significant electoral manipulation. By introducing and validating CRW as an unsupervised mitigation strategy, this work takes a step toward safeguarding VAA integrity, ensuring that voter-candidate recommendations reflect genuine policy preferences rather than exploitation of structural vulnerabilities.

ETHICAL STATEMENT

The dataset was collected and anonymized by Politools in compliance with the new Swiss Federal Act on Data Protection (nFADP), the Telecommunications Act (TCA), and applicable data protection regulations. We adhere strictly to the platform’s terms of use, in particular by not reporting results attributable to individuals. The dataset remains private, but may be shared with researchers by Politools [19] upon request.

CREDIT AUTHOR STATEMENT

Liv Weiss: Investigation, Software, Validation, Visualization, Writing - Original Draft. *Damien Berriaud:* Conceptualization, Methodology, Supervision, Visualization, Writing - Original Draft, Review & Editing. *Frederic Berdoz:* Conceptualization, Methodology, Supervision. *Roger Wattenhofer:* Supervision, Funding acquisition.

REFERENCES

- [1] D. Balduzzi, K. Tuyls, J. Pérolat, and T. Graepel. 2018. Re-evaluating Evaluation. In *Advances in Neural Information Processing Systems (NeurIPS)*, Vol. 31. 3272–3283.

- [2] F. Berdoz, D. Brunner, Y. Vonlanthen, and R. Wattenhofer. 2025. Recommender Systems for Democracy: Toward Adversarial Robustness in Voting Advice Applications. In *International Joint Conference on Artificial Intelligence (IJCAI)*.
- [3] D. Berriaud and R. Wattenhofer. 2025. Clone-Robust Weights in Metric Spaces: Handling Redundancy Bias for Benchmark Aggregation. arXiv:2502.03576
- [4] D. Berriaud and R. Wattenhofer. 2026. Breaking the Illusion of Artificial Consensus: Clone-Robust Weighting for Arbitrary Metric Spaces. arXiv:2602.24024
- [5] J. Chen, S. Xiao, P. Zhang, K. Luo, D. Lian, and Z. Liu. 2024. M3-Embedding: Multi-Linguality, Multi-Functionality, Multi-Granularity Text Embeddings Through Self-Knowledge Distillation. In *Findings of the Association for Computational Linguistics: ACL 2024*. 2318–2335. doi:10.18653/v1/2024.findings-acl.137
- [6] V. Conitzer and M. Yokoo. 2010. Using Mechanism Design to Prevent False-Name Manipulations. *AI Magazine* 31, 4 (2010), 65–78.
- [7] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova. 2019. BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding. arXiv:1810.04805
- [8] V. Isotalo. 2021. Improving Candidate-Based Voting Advice Application Design: The Case of Finland. *Informaatietutkimus* 40, 3 (2021), 85–109. doi:10.23978/inf.107215
- [9] Ozan Kuru and Josh Pasek. 2016. Improving social media measurement in surveys: Avoiding acquiescence bias in Facebook research. *Computers in Human Behavior* 57 (April 2016), 82–92. doi:10.1016/j.chb.2015.12.008
- [10] A. Ladner, J. Fivaz, and J. Pianzola. 2012. Voting Advice Applications and Party Choice: Evidence from SmartVote Users in Switzerland. *International Journal of Electronic Governance* 5, 3–4 (2012), 367–387. doi:10.1504/IJEG.2012.051303
- [11] A. Ladner and J. Pianzola. 2010. Do Voting Advice Applications Have an Effect on Electoral Participation and Voter Turnout? Evidence from the 2007 Swiss Federal Elections. In *IFIP International Conference on Electronic Participation (ePart)*.
- [12] B. Lee, R. Aiyappa, Y.-Y. Ahn, H. Kwak, and J. An. 2025. A Semantic Embedding Space Based on Large Language Models for Modelling Human Beliefs. *Nature Human Behaviour* 9, 9 (2025), 1928–1940.
- [13] Z. Li, X. Zhang, Y. Zhang, D. Long, P. Xie, and M. Zhang. 2023. Towards General Text Embeddings with Multi-stage Contrastive Learning. arXiv:2308.03281
- [14] T. Louwerse and M. Rosema. 2014. The Design Effects of Voting Advice Applications: Comparing Methods of Calculating Matches. *Acta Politica* 49 (2014), 286–312.
- [15] L. Marris, S. Liu, I. Gemp, G. Piliouras, and M. Lanctot. 2025. Deviation Ratings: A General, Clone-Invariant Rating Method. arXiv:2502.11645
- [16] S. Munzert and S. Ramirez-Ruiz. 2021. Meta-Analysis of the Effects of Voting Advice Applications. *Political Communication* 38, 6 (2021), 691–706. doi:10.1080/10584609.2020.1843572
- [17] Z. Nussbaum, J. X. Morris, B. Duderstadt, and A. Mulyar. 2024. Nomic Embed: Training a Reproducible Long Context Text Embedder. arXiv:2402.01613
- [18] OpenAI. 2024. GPT-4o System Card. arXiv:2410.21276
- [19] Politools. 2024. SmartVote. smartvote.ch Accessed January 2025.
- [20] A. D. Procaccia, B. Schiffer, and S. Zhang. 2025. Clone-Robust AI Alignment. In *Proceedings of the 42nd International Conference on Machine Learning (ICML)*.
- [21] N. Reimers and I. Gurevych. 2019. Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks. arXiv:1908.10084
- [22] S. Sturua, I. Mohr, M. K. Akram, M. Günther, B. Wang, M. Krimmel, F. Wang, G. Mastrapas, A. Koukounas, N. Wang, and H. Xiao. 2024. jina-embeddings-v3: Multilingual Embeddings With Task LoRA. arXiv:2409.10173
- [23] L. Terán. 2020. Voting Advice Applications. In *Dynamic Profiles for Voting Advice Applications*. Springer, 15–26. doi:10.1007/978-3-030-24090-5_2
- [24] T. N. Tideman. 1987. Independence of Clones as a Criterion for Voting Rules. *Social Choice and Welfare* 4, 3 (1987), 185–206. doi:10.1007/BF00433944
- [25] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. Kaiser, and I. Polosukhin. 2017. Attention Is All You Need. arXiv:1706.03762
- [26] S. Walgrave, M. Nuytemans, and K. Pepermans. 2009. Voting Aid Applications and the Effect of Statement Selection. *West European Politics* 32, 6 (2009), 1161–1180. doi:10.1080/01402380903230637
- [27] L. Wang, N. Yang, X. Huang, B. Jiao, L. Yang, D. Jiang, R. Majumder, and F. Wei. 2022. Text Embeddings by Weakly-Supervised Contrastive Pre-training. arXiv:2212.03533
- [28] L. Wang, N. Yang, X. Huang, L. Yang, R. Majumder, and F. Wei. 2024. Improving Text Embeddings with Large Language Models. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (ACL)*. 11897–11916. doi:10.18653/v1/2024.acl-long.642
- [29] Y. Zhang, M. Li, D. Long, X. Zhang, H. Lin, B. Yang, P. Xie, A. Yang, D. Liu, J. Lin, F. Huang, and J. Zhou. 2025. Qwen3 Embedding: Advancing Text Embedding and Reranking Through Foundation Models. arXiv:2506.05176

VIII. SWISS POLITICAL CONTEXT & SMARTVOTE DATASET

This section provides additional details on the Swiss electoral context, the SmartVote platform, and the dataset underlying our empirical analysis.

A. SmartVote and the Swiss Electoral Context

SmartVote is Switzerland’s primary Voting Advice Application, operated by the non-profit association Politools [19]. In the 2023 Swiss National Council elections, it was used by up to 20% of eligible voters [19], which makes a high-stakes platform for studying adversarial robustness in voter-candidate recommendation systems. Roughly 85% of all registered candidates completed the SmartVote questionnaire [19].

The Swiss National Council is constituted of 200 seats distributed proportionally across the 26 cantons, with each canton allocated seats roughly in proportion to its population. Elections are held every four years, and voters may only cast votes for candidates registered in their own canton [2].

Candidates standing for election are required to form *lists*, which are party- or coalition-specific slates of candidates. The number of candidates on a list corresponds to the number of seats available in the canton. Voters are presented with the option of either voting for a predefined list or composing their own custom list by selecting individual candidates across different lists. Voters may also give additional weight to a specific candidate by placing their name twice on the list, effectively granting them two votes and boosting their position within the list [2]. The total votes accumulated across all lists associated with a party determines how many seats that party wins, and seats are then assigned to candidates in order of their ranking on the list, from top to bottom. A party’s overall strength therefore depends on how frequently its candidates appear in voters’ selections, while the internal ranking on the list determines which of those candidates are ultimately elected.

The Swiss political landscape is defined by a highly stable multi-party system. Nationally, the electorate is dominated by six major parties spanning the political spectrum: the right-wing conservative Swiss People’s Party (SVP), the left-wing Social Democratic Party (SP), the centre-right FDP. The Liberals (FDP), the centrist The Centre (Centre), the left-leaning Green Party (Green), and the centrist Green Liberal Party (GLP). Together, these parties capture the vast majority of the national vote and provide the primary ideological axes along which VAAs position candidates and voters.

All experiments in this work use the SmartVote dataset for the 2023 National Council elections in Canton Zurich, collected and anonymized by Politools in accordance with applicable Swiss data protection law.

B. Questionnaire Structure

The SmartVote questionnaire is newly created for each election cycle. For the 2023 elections, it consists of 75 questions

divided into three types: 60 policy questions with answer options $A_t = \{0, 25, 75, 100\}$, 7 value questions with $A_t = \{0, 17, 33, 50, 67, 83, 100\}$, and 8 budget questions with $A_t = \{0, 25, 50, 75, 100\}$. All questions accept a voter-assigned importance weight from $W_t = \{0, 0.5, 1, 2\}$, where 1 is the default for answered questions and 0 is assigned automatically to skipped questions [2].

Questions are assigned to one of 14 topic categories by SmartVote: *Federal budget*, *Energy & transport*, *Values*, *Democracy*, *Media & Digitalization*, *Nature conservation*, *Welfare state & family*, *Economy & labour*, *Foreign trade & foreign policy*, *Health*, *Security & military*, *Finances & taxes*, *Immigration & integration*, *Society & ethics*, and *Education*. The number of questions per category ranges from 3 (Education) to 8 (Federal budget). The mini questionnaire covers 12 of the 14 categories, with *Federal budget* and *Values* absent entirely, as all of their questions appear only in the full questionnaire.

In addition to the full 75-question survey, SmartVote offers a 30-question *rapide* version, hereafter referred to as the mini questionnaire, curated by SmartVote experts as a representative subset of the full questionnaire. The mini questionnaire covers 12 of the 14 topic categories, with the *Values* and *Federal budget* questions only occurring in the full questionnaire.

C. Recommendation Calculation

The Canton Zurich dataset contains 41,658 voters and 1,029 candidates registered on SmartVote. For each voter v_i , SmartVote computes a distance to every candidate c_j using a weighted L2 metric and ranks candidates by proximity. In its original formulation, SmartVote applies importance weights as follows:

$$d_{\text{SmartVote}}(\mathbf{v}_i, \mathbf{w}_i, \mathbf{c}_j) = \sqrt{\sum_{t=1}^{N_q} (w_{i,t}(v_{i,t} - c_{j,t}))^2} \quad (1)$$

where $w_{i,t} \in W_t = \{0, 0.5, 1, 2\}$ is the importance weight assigned by voter v_i to question q_t , with 1 as the default for answered questions and 0 for skipped questions. The top 36 candidates—corresponding to the number of National Council seats allocated to Canton Zurich—are prominently displayed to the voter.

D. Data Cleaning

The raw dataset was preprocessed following the procedure described by Berdoz et al. [2]: corrupt entries lacking essential metadata were removed, voters who answered fewer than 15 questions were excluded ($< 1.3\%$ of the dataset), and duplicate entries were resolved by retaining only the most recent recommendation with the highest number of answered questions per unique voter ID. Recommendations submitted before the questionnaire went live or after the election date were filtered out, as were responses exhibiting random clicking behaviour (more than 14 consecutive identical answers, $\approx 0.1\%$). On

the candidate side, candidates who did not participate in SmartVote were removed, and youth party affiliations were merged with their parent parties (e.g., JUSO with SP, JSVP with SVP).

E. Usage of Importance Weights

In the whole 2023 Swiss National Council elections dataset, only 45.8% of all voters (86,138) entered at least one value different from the default weight of one. Among voters that modified at least one value, the average number of importance weights different from one is 11.35, representing only 15.1% of all questions.

IX. ADDITIONAL METHODOLOGICAL DETAILS

This section provides additional methodological details underlying the experimental setup, including clone generation, embedding model instructions, and procedures for model and hyperparameter selection.

A. Clone Generation: Prompts and Examples

All synthetic clones are generated using GPT-4o (gpt-4o-2024-08-06) with temperature 0.8 and structured JSON output. Each prompt follows a shared template:

```
You are helping generate test data for
a Swiss political questionnaire.

Original question:
"{question_text}"

Task: {type-specific instruction}

Return a JSON object with a single key
"paraphrase" containing your result.
Example: {"paraphrase": "Your
paraphrase here."}
```

When multiple clones of the same type are needed for a single question, all previously generated paraphrases of that type are appended to the prompt with the instruction: “The following paraphrases of this type have already been generated. You MUST produce a DIFFERENT one.” For the hard paraphrase and hard negation types, up to three previously generated easy paraphrases are additionally included as negative exemplars with the prefix: “For reference, these are EASY paraphrases (similar wording). Your hard paraphrase must be MUCH more different than these,” followed by the supplementary rules shown below.

a) Easy Paraphrase:

Rephrase the question using similar but not identical wording and a slightly different sentence structure. The meaning must be exactly the same. Some words may overlap with the original.

b) Hard Paraphrase:

Rephrase the question using COMPLETELY DIFFERENT words and sentence structure. The meaning must be identical, but you MUST NOT share any significant content words with the original. Use synonyms, different grammatical forms, or

circumlocutions. You may change the grammatical form (e.g. question to statement or vice versa).

For example: if the original says ‘taxes’, use ‘fiscal levies’ or ‘government revenue’. If it says ‘legalized’, use ‘permitted by law’ or ‘decriminalized’. If it says ‘retirement age’, use ‘the point at which workers leave the workforce’ or ‘the threshold for ending one’s professional career’.

IMPORTANT RULES:

- You MUST NOT reuse any significant content words from the original question.
- Nouns, verbs, and adjectives that carry the topic meaning must ALL be replaced with synonyms or circumlocutions.
- If you find yourself writing a word that appears in the original, STOP and find an alternative.
- The result should look like it was written by someone who never saw the original, describing the same concept from scratch.

c) Easy Negation :

First, create an easy paraphrase of the original (similar wording, slightly different structure). Then negate that paraphrase so that someone who agrees with the original would disagree with your version. Return only the final negated paraphrase.

d) Hard Negation :

First, create a hard paraphrase of the original -- using COMPLETELY DIFFERENT vocabulary and sentence structure, with NO significant content words from the original. Then negate that hard paraphrase so that someone who agrees with the original would disagree with your version. Return only the final negated paraphrase.

For example: if the original says ‘taxes’, the hard paraphrase should use ‘fiscal levies’ or ‘government revenue’ instead. Then negate that rephrased version.

IMPORTANT RULES:

- You MUST NOT reuse any significant content words from the original question.
- Nouns, verbs, and adjectives that carry the topic meaning must ALL be replaced with synonyms or circumlocutions.
- If you find yourself writing a word that appears in the original, STOP and find an alternative.
- The result should look like it was written by someone who never saw the

original, describing the same concept from scratch.

The generated clones were manually inspected to verify that they address the same policy issue as the original question. Table IV finally shows an example of all four clone types for a representative question.

B. Details regarding Model and α Selection

This section covers additional methodological details regarding model and α selection left out of the main body for the lack of sufficient.

a) Embedding Model Instructions: Four models accept task-specific instructions prepended to the input text. The instructions used are shown in Table V. Models with no prefix receive raw question text. Instruction-aware models receive either a prefix template or a task keyword passed as an encode argument.

To ensure a consistent distance metric for CRW when using an asymmetric approach, the distance between any two questions q_i and q_j is symmetrized by averaging the two embedding distances:

$$d^*(q_i, q_j) = \frac{d(q_i, q_j) + d(q_j, q_i)}{2}$$

where $d(q_i, q_j)$ represents the distance from q_i to q_j under the model’s specific instruction set.

b) Most Impactful Question Selection: To identify the most impactful questions, we craft for each question $q \in \mathcal{Q}$ a manipulated questionnaire $\mathcal{Q}' = \mathcal{Q} \cup \{q'_1, \dots, q'_4\}$, where each q'_s is a 0-behavioral 0-embedding clone of q .

Let τ denote the Kendall ranking distance, defined by

$$\tau(r_i, r'_i) = \frac{|\mathcal{C}^+| - |\mathcal{C}^-|}{\binom{N_c}{2}} \quad (2)$$

For every distortion measure $m \in \{J_{36}, \rho_S, \tau\}$, we then compute the average recommendation distortion across voters, and rank questions by the distortion of their corresponding attack set for each of the metric individually. We select the five questions with the lowest mean rank across all three metrics. Table VI displays the resulting five most impactful questions.

c) Selection of optimal radii α^ :* To select the best radius α^* for each model, we compute for each metric $m \in \{J_{36}, \rho_S, \tau\}$ the average recommendation distortion across all voters and attack settings. We then aggregate these three measures of distortion within a composite score that ensures scale invariance across measures, i.e.,

$$\text{Score}(\text{model}, \alpha) = \sum_{m \in \{J_{36}, \rho_S, \tau\}} \frac{m_{\text{model}}^{\text{CRW}}(\alpha) - \min m_{\text{model}}^{\text{CRW}}}{\max m_{\text{model}}^{\text{CRW}} - \min m_{\text{model}}^{\text{CRW}}}, \quad (3)$$

where $m_{\text{model}}^{\text{CRW}}(\alpha)$ denotes the average distortion across voters and attacks according to measure m at radius α , and the min and max are taken over all values of α , for a fixed model.

For each model, we then select the value α^* that maximizes the function $\text{Score}(\text{model}, \cdot)$.

X. ADDITIONAL RESULTS

This section provides additional experimental results, covering robustness measures agreement, per-question distortion analyses, model and hyperparameter selection, and robustness checks on the full questionnaire.

A. Agreement between Robustness Measures

To justify reporting Jaccard similarity as the sole robustness measure throughout the main text, we verify that all three ranking measures $m \in \{J_{36}, \rho_S, \tau\}$, c.f. Equation (2), agree on the relative severity of distortion across questions and clone intensities. For each of the 75 questions cloned at 10 intensities ($\ell \in \{1, 2, 3, 4, 5, 7, 10, 15, 20, 30\}$) with identical clones under uniform weights, we compute the mean distortion across all voters under each metric.

Figure 5 shows the pairwise relationships across all 750 (question, k) combinations. The three measures pairs exhibit near-perfect rank agreement: Spearman correlations range from $\rho_S = 0.965$ (Jaccard vs. Spearman’s ρ_S) to $\rho_S = 0.994$ (Spearman’s ρ_S vs. Kendall’s τ). The metrics rank questions in effectively the same order at every clone intensity, and the agreement holds across all subsequent experiments—the metric used does not affect which model is selected, which α is optimal, or which questions are identified as most vulnerable.

Given this strong agreement, we adopt Jaccard as the primary measure for two reasons. First, it is the most directly relevant to electoral outcomes. SmartVote displays the top 36 candidates to each voter, corresponding to the number of National Council seats in Canton Zurich. Voters who fully adopt this recommended list determine the party seat allocation; Jaccard over the top-36 set therefore measures exactly what matters—whether the same candidates appear in a voter’s recommendation, regardless of their internal ordering. Spearman’s ρ_S and Kendall’s τ , by contrast, are sensitive to rank permutations deep in the list that have no practical consequence for voter behavior or party visibility. Second, Jaccard has a wider dynamic range on the distortions we study: at $k = 4$ identical clones, Jaccard values span 0.657–0.928 across questions, while Spearman’s ρ_S compresses into 0.92–0.99 and Kendall’s τ into 0.80–0.97. The greater spread makes Jaccard more informative for discriminating between models, α values, and attack settings.

B. Vulnerability Assessment: Per-Question Distortion

Figure 7 displays the per-question distortion obtained after injecting $\ell = 4$ exact copies of a question.

C. Additional Results for the Model and α Selection

We present and discuss in these section a few additional results regarding the model selection step.

Condition	Question text
Original	Do you support an increase in the retirement age (e.g., to 67)?
Easy paraphrase	Are you in favor of raising the retirement age, such as to 67 years?
Hard paraphrase	Are you in favor of extending the age limit for concluding one’s career to, say, 67 years?
Negation (easy)	Are you against raising the retirement age to 67?
Negation (hard)	Would you oppose keeping the current age limit for receiving pension benefits, such as not raising it to 67?

Table IV: Example clones for Q32214. The mixed condition combines one clone of each of the other four types.

Model	Input prefix	Instruction
SBERT	—	—
E5	query: {text}	—
E5-Asym	query: {text} / passage: {text}	—
E5-Instruct	Instruction: {instr}\nQuery: {text}	<i>Identify the political topic discussed in the following question, regardless of the stance or position taken.</i>
E5-Asym-Instruct	See E5-Instruct (asymmetric encoding)	See E5-Instruct
BGE-M3	—	—
GTE	—	—
Nomic v2	clustering: {text}	—
Jina v3	Raw text. Task via encode kwarg	embedding_task = "separation"
Qwen3	Raw text. Instruction via prompt encode kwarg	<i>Classify the following political question by its topic or policy area, ignoring whether the question is for or against the policy.</i>

Table V: Instructions used for each Embedding Model. Empty entries signal that the Model directly receive the raw question text.

Rank	Q ID	Question	Category
1	32214	Do you support an increase in the retirement age (e.g., to 67)?	Welfare state & family
2	32234	Should a third official gender be introduced alongside “female” and “male”?	Society & ethics
3	32261	A popular initiative aims to reduce television and radio fees (CHF 200 per household, exemption for businesses). Do you support this initiative?	Democracy, Media & Digitalization
4	32228	Should the conditions for naturalization be relaxed (e.g., shorter residence period)?	Immigration & integration
5	32222	Should insured persons contribute more to health care costs (e.g., increase the minimum deductible)?	Health

Table VI: The five most impactful questions, selected by lowest composite impact rank across J_{36} , ρ_S , and τ . All five questions happen to be included in the mini questionnaire (rapide).

a) Clone-Type Variation.: Figure 7 displays the rank of each model within each clone condition, ordered by difficulty. The ranking reveals E5-Instruct’s broad dominance: it ranks first across four of the five conditions, encompassing both paraphrases and negations at both difficulty levels. The notable exception is the mixed condition, where E5-Instruct drops to third, outperformed by Jina-v3 and Qwen3. This presents a curious anomaly: a model that excels at detecting individual clone strategies in isolation struggles relatively when those strategies are deployed simultaneously. Beneath the top tier, there is significant rank volatility. Qwen3, for instance, overtakes GTE and BGE-M3 on easy negations, but plummets to 8th on hard negations, where GTE and BGE-M3 recover

their standings. Finally, E5-Asym-Instruct remains anchored at the bottom, ranking 10th on four of the five conditions. It escapes last place only on hard negations, where 4 other models perform even worse.

b) Selected Configuration.: Driven by the composite ranking procedure, the selected defense configuration is **E5-Instruct** paired with a CRW radius of $\alpha^* = 0.4$. While E5-Instruct’s performance actually peaks at $\alpha = 0.3$ under certain conditions (most notably the easy paraphrase condition) expanding the radius to 0.4 maximizes the aggregate composite score across the full spectrum of difficulty tiers. The rationale for favoring this slightly larger radius is rooted in the threat model: a strict threshold of $\alpha = 0.3$ operates too conservatively,

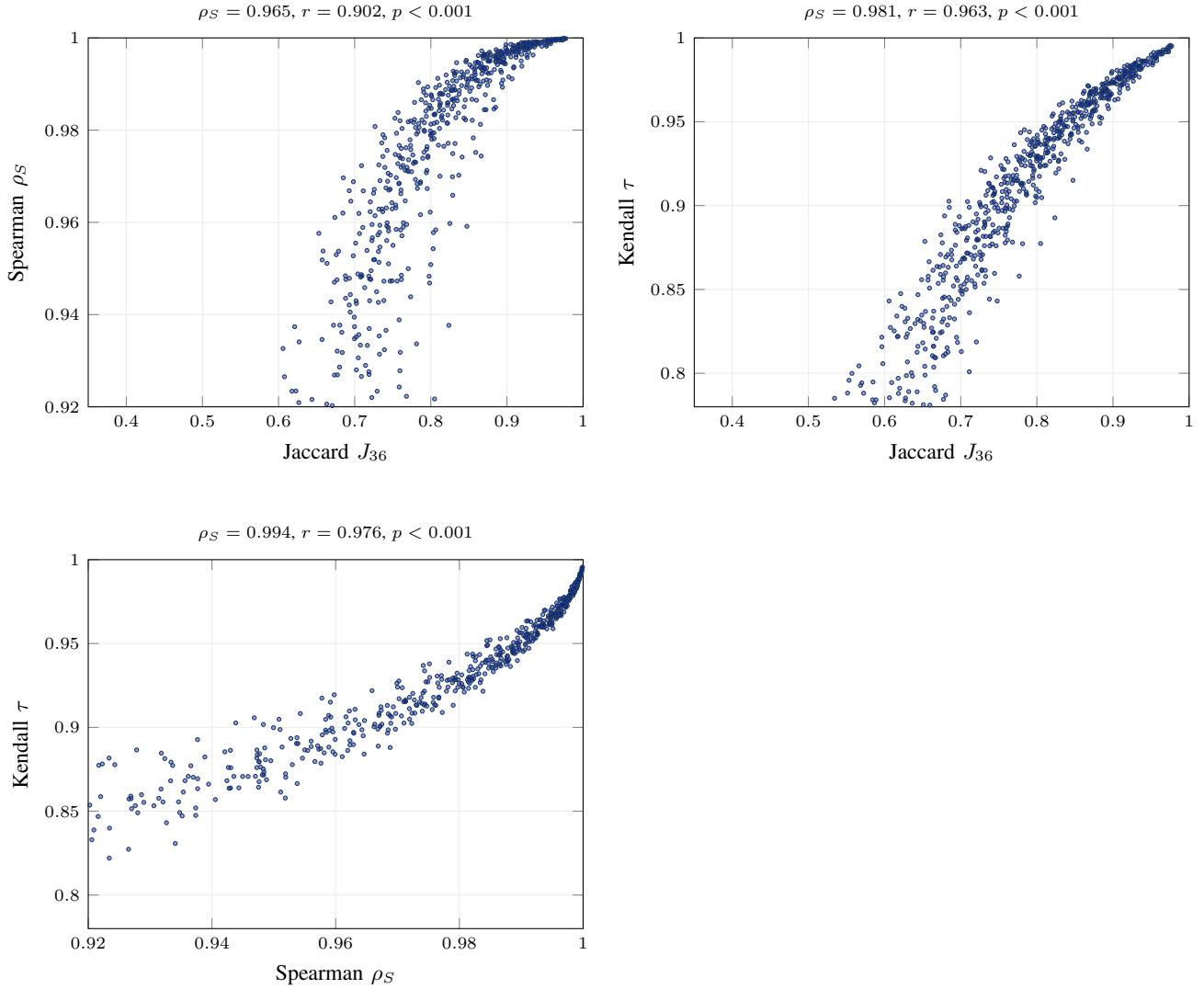


Figure 5: Pairwise agreement between Jaccard similarity (J_{36}), Spearman’s ρ_S , and Kendall’s τ across all 75 questions and 10 clone counts ($k \in \{1, \dots, 30\}$, $n = 750$ points per panel). Each dot represents one (question, k) combination under identical cloning with uniform weights. All three metric pairs show near-perfect rank agreement ($\rho_S > 0.934$, $p < 0.001$), justifying the use of Jaccard as the primary metric throughout this thesis. Spearman and Kendall are the most tightly coupled ($\rho_S = 0.988$), while Jaccard shows slightly more spread relative to both rank-based metrics due to its sensitivity to the top-36 boundary.

successfully capturing obvious lexical duplicates but risking the exclusion of more complex, structurally distinct clones (such as hard paraphrases and hard negations) that naturally sit at slightly higher embedding distances. Furthermore, because our parameter sweep evaluated discrete intervals, the “true” continuous optimum likely resides somewhere between 0.3 and 0.4. However, as this framework serves primarily as a proof of concept, pinpointing a hyper-exact scalar value is less critical than establishing a stable, robust threshold capable of generalizing across diverse attack vectors. This configuration is therefore utilized in all subsequent experiments unless otherwise stated. The underlying structural mechanisms driving this performance—and specifically why E5-Instruct outperforms models that more closely mimic ground-truth

correlation metrics—are dissected in Appendix XI.

D. Full Questionnaire Robustness

We next discuss a few additional results regarding the robustness evaluation of the selected defense configuration on the full questionnaire.

a) Alpha Stability.: The optimal aggregate radius identified across all 75 questions and five clone conditions remains $\alpha^* = 0.4$ (Figure 8), matching exactly the compromise value selected during the model selection phase. While easier clones still exhibit a marginal peak at $\alpha = 0.3$, the $\alpha = 0.4$ threshold consistently provides the best aggregate performance when harder paraphrases and negations are included. This consistency confirms that the radius derived from the restricted 5-question target set was not an artifact of that specific subset,

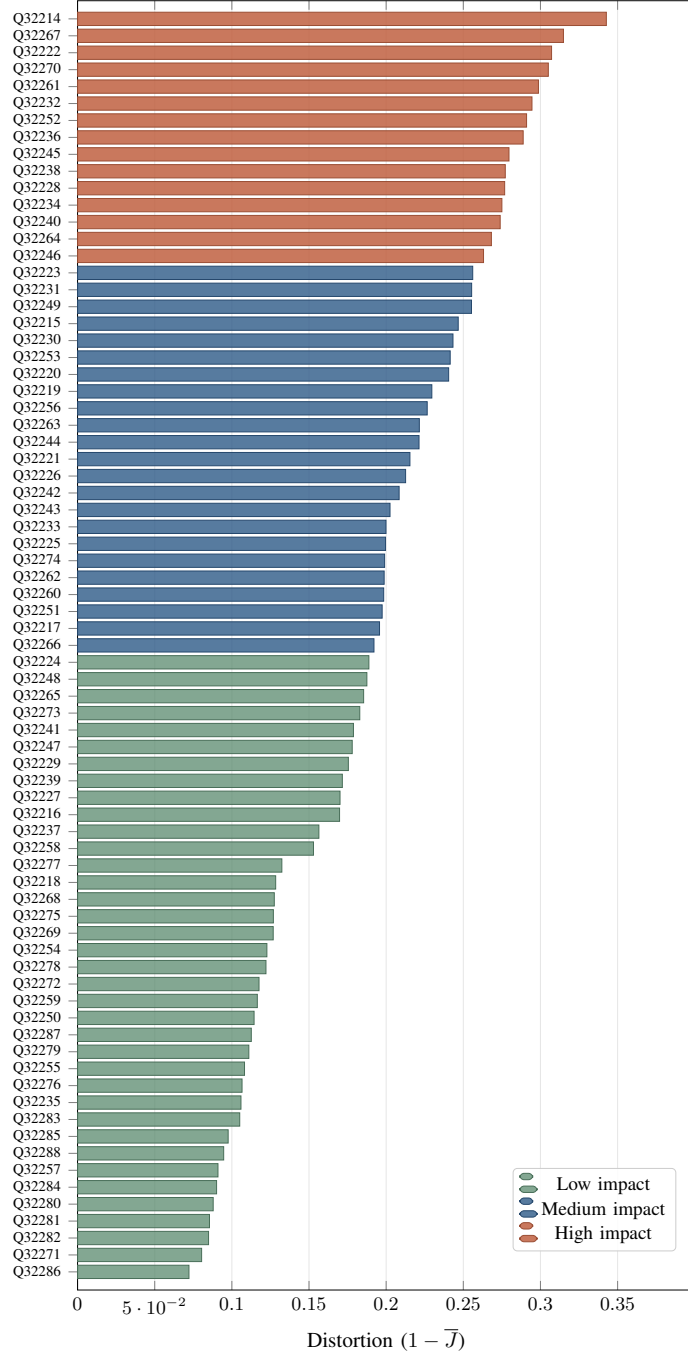


Figure 6: Per-question distortion at $\ell = 4$ clones, measured as $1 - \bar{J}$ (mean Jaccard distance from the unmanipulated ranking). Questions are ordered by level of distortion; colors indicate distortion tercile.

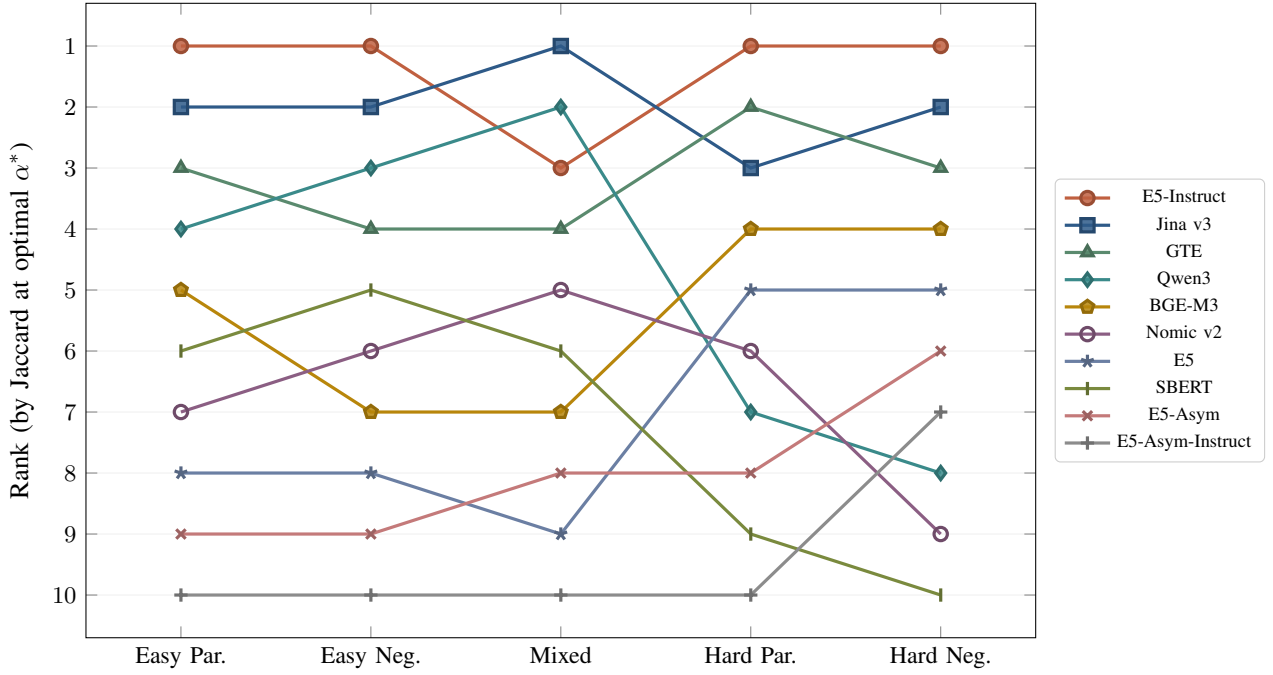


Figure 7: Model ranking by clone type, based on maximum Jaccard similarity at each model’s optimal α^* . Clone conditions are ordered by approximate difficulty (left to right). The top two models (E5-Instruct, Jina v3) maintain near-constant rankings across all conditions, while mid-tier models exhibit considerable reshuffling: Qwen3 ranks 2nd on the mixed condition but drops to 8th on hard negations, and E5-Asym-Instruct ranks last on four conditions but climbs to 7th on hard negations.

but rather reflects a stable structural property of E5-Instruct’s embedding space across diverse topics.

b) Correlation metric ceiling.: Figure 9 shows the α sweep for the correlation-based distance metric $d_{\text{corr}}(q, q') = \arccos(|\rho(q, q')|)$, evaluated using 0-embedding 0 behavioral clones across all 75 questions. Since these clones receive identical response vectors to their originals by construction, the metric assigns them distance exactly zero and is therefore clone-type invariant: a single clone condition suffices.

The correlation metric achieves near-perfect mean Jaccard of 0.998 across a wide α plateau of $[0.01, 1.50]$, with 71 of 75 questions exceeding 0.99 at $\alpha = 1.50$. The small residual imperfection at low α values (mean Jaccard = 0.992 for $\alpha \leq 0.60$) is an artifact of floating-point imprecision: several question pairs are assigned identical distances, causing non-deterministic tie-breaking in the equivalence class computation that marginally perturbs CRW weights. Theoretically, Jaccard should be exactly 1.0 for all α below the minimum distance between two distinct questions $d_{\min} \approx 0.68$, since clones are assigned distance exactly zero and no non-clone pair fall within that radius.

The gap between E5-Instruct (mean Jaccard 0.870 at $\alpha = 0.4$, averaged over all clone types) and this ceiling (0.998) suggests that the limiting factor is the embedding model’s ability to cluster clones in text space, not the CRW framework itself. The relationship between the two distance structures is examined in Appendix XI-C.

E. Targeted Party Manipulation

This section presents additional detailed results for the targeted manipulation attacks described in the main body of the paper.

a) Unmitigated attack impact.: Without CRW, targeted cloning consistently distorts party visibility across all six parties. Under the worst-case (see Table VII), visibility gains range from +1.37pp (SVP) to +4.76pp (Centre). Even the realistic attack (see Table III for an overview), consisting of just five injected questions, produces meaningful distortions for all six parties. Gains range from +0.47pp (SVP) to +1.83pp (Centre). The Centre party benefits most in both scenarios, followed by SP and GLP. Because visibility is inherently zero-sum, these artificial gains come directly at the expense of other parties. However, the collateral impact is not distributed evenly across the political spectrum. When the Centre or left-leaning parties are targeted, they largely siphon visibility from one another, while the centre-right and right-wing parties (FDP, SVP) remain relatively insulated. For example, cloning Centre’s top questions boosts its visibility by +4.76pp while costing SP −2.35pp and Green −2.39pp, yet leaves FDP and SVP largely unaffected (+0.19pp and +0.32pp, respectively).

b) CRW natural drift.: Applying CRW to the original unmanipulated questionnaire produces negligible shifts in party visibility for all six parties (maximum |drift| = 0.12pp for Centre). This reflects the good design of the SmartVote questionnaire, whose questions are well-separated in embedding space and present little semantic redundancy—leaving CRW

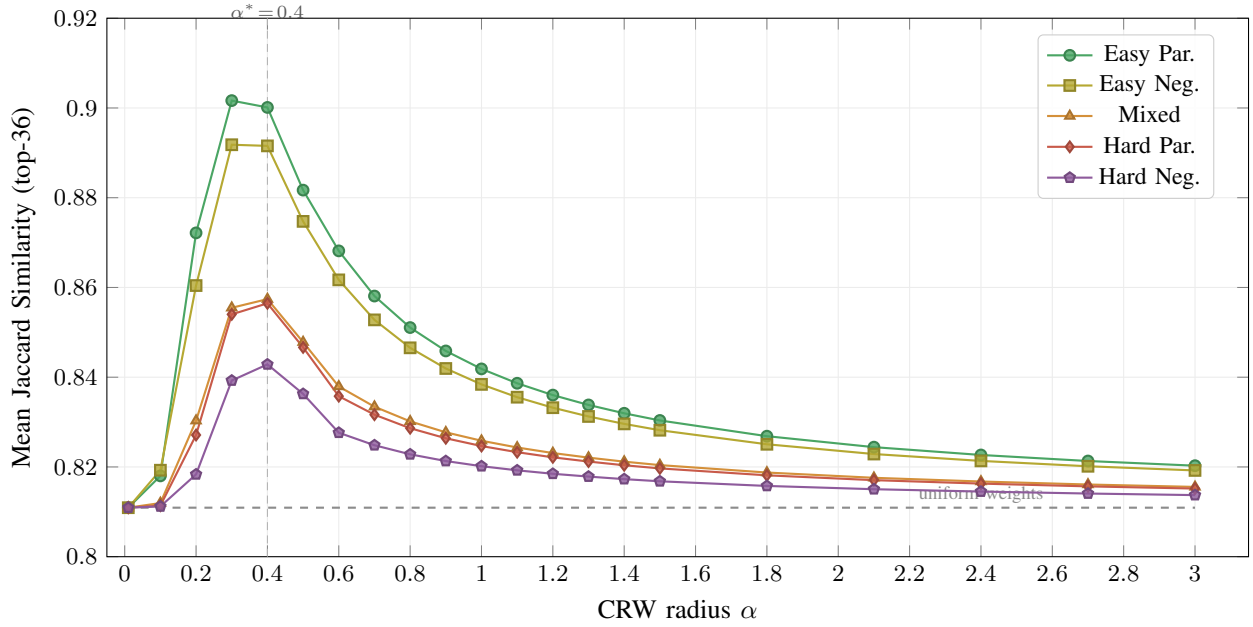


Figure 8: Mean Jaccard similarity as a function of CRW radius α across all 75 questions, by clone type (E5-Instruct). The dashed horizontal line marks the uniform-weight baseline. All clone types peak near $\alpha = 0.4$ (dashed vertical line): easy clones peak marginally at $\alpha = 0.3$ while harder clones peak at $\alpha = 0.4$, making $\alpha^* = 0.4$ the best compromise across conditions. Beyond $\alpha \approx 0.6$, performance degrades as CRW begins collapsing genuinely distinct questions into shared equivalence classes.

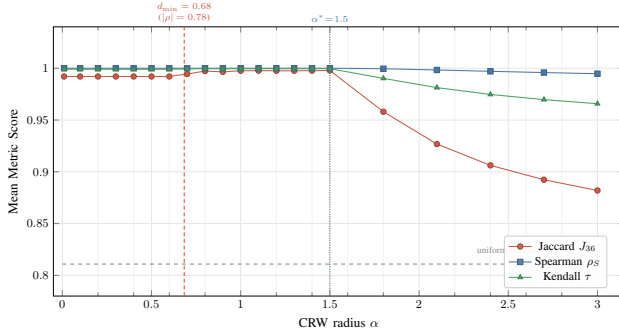


Figure 9: CRW performance using the correlation-based distance metric $d_{\text{corr}}(q, q') = \arccos(|\rho(q, q')|)$ with identical clones, averaged over all 75 questions. CRW trivially achieves near-perfect correction ($J > 0.99$) across a wide α range of 0.01–1.5. The dashed line marks $d_{\text{min}} = 0.68$, the distance of the closest non-clone question pair ($|\rho| = 0.78$). At $\alpha = 1.5$ performance starts drastically degrading. The residual imperfection below d_{min} (mean Jaccard = 0.992 for $\alpha \leq 0.60$) is an artefact of floating-point imprecision.

with little to correct in the absence of adversarial manipulation. We further discuss the geometry of the SmartVote dataset in Appendix XI-C.

XI. COMPLEMENTARY ANALYSES

This section presents three complementary analyses. Section XI-A investigates which factors best explain why certain questions are more potent vehicles for redundancy attacks.

Attack	Visibility Change (pp)					
	Centre	SP	GLP	Green	FDP	SVP
Centre	+4.76	-2.35	-1.33	-2.39	+0.19	+0.32
SP	-1.20	+4.53	-1.85	-0.57	-0.51	-0.26
GLP	-0.77	-1.40	+4.16	+0.64	+0.17	+0.17
Green	-1.45	+1.19	-1.63	+2.20	-0.58	-0.37
FDP	-0.88	-1.23	-0.83	-0.54	+1.44	+0.71
SVP	-0.84	-0.17	+0.23	-0.63	+1.27	+1.37

Table VII: Cross-party visibility impact of worst-case attacks ($4\times$ mixed clones). Each row corresponds to an attack designed to benefit the row’s party, while columns show the resulting visibility change (in percentage points) for all six major parties under uniform weights. Diagonal entries (bold) are the targeted party’s gain. Off-diagonal entries reveal collateral effects: an attack benefiting one party systematically harms others, though the magnitude and distribution of harm varies across the political spectrum.

Section XI-B examines the structural properties of sentence embedding models that determine their effectiveness when combined with CRW. Finally, Section XI-C studies the global geometry of the SmartVote dataset under both embedding and behavioral correlation metrics.

A. Predictors of Question Duplication Impact

Figure 7 revealed that clone impact is highly non-uniform across the questionnaire, with recommendation distortion ranging from 0.072 to 0.343 under identical attack intensities ($k = 4$). To map these dynamics, we examine which question-level

properties best predict adversarial impact. Figure 10 displays scatter plots for six candidate predictors, and Table VIII reports the corresponding Spearman rank correlations.

a) Answer variance is the dominant predictor.: Voter answer variance is the strongest individual predictor of clone impact ($\rho_S = +0.806$, $p < 0.001$), followed by the product of voter and candidate variance ($\rho_S = +0.782$) and candidate variance alone ($\rho_S = +0.675$). The mechanism is direct: cloning q ℓ -times multiplies its contribution by $\ell+1$. Questions where respondents spread across the full answer spectrum produce large squared differences, so the multiplicative amplification from cloning translates into large absolute distortion. Questions where most respondents answer similarly produce small squared differences, and cloning them has limited effect regardless of ℓ .

b) Skip rate is a strong negative predictor.: The fraction of voters who skip a question (voter NaN%) is a strong negative predictor ($\rho_S = -0.654$, $p < 0.001$): skipped questions receive weight zero and therefore do not contribute to the voter-candidate distance, making cloning them ineffective.

Question position in the survey is also negatively correlated with impact ($\rho_S = -0.556$, $p < 0.001$), possibly serving as a proxy for skip rate: questions appearing later in the 75-question survey are skipped more frequently due to respondent fatigue [2]. Voter variance and skip rate are themselves correlated: questions on which voters disagree strongly tend to be answered rather than skipped, so these predictors are not fully independent.

c) Inter-question correlation does not predict impact.: Theoretically, one would expect a strong negative relationship between a question’s baseline correlation with other survey items and its adversarial clone impact. If a question is already highly correlated with multiple existing items, its underlying policy dimension is already heavily weighted in the baseline L2 distance computation. Adding a synthetic clone to a saturated dimension should yield diminishing marginal returns on the final candidate ranking. Because the policy axis is already represented by a cluster of ℓ correlated questions, injecting one additional clone merely shifts its cumulative weight to $\ell+1$, a proportionally minor increase compared to abruptly doubling the impact of a strictly isolated question.

Empirically, the data provides a consistent but statistically insignificant directional signal for this hypothesis. None of the eight inter-question correlation metrics tested reach strict statistical significance (Table VIII). The strongest metric is the number of correlated neighbors ($\rho_S = -0.161$, $p = 0.169$), the weakest is the sum of squared correlations on the candidate side ($\rho_S = -0.040$, $p = 0.735$). It is important to note, however, that this lack of significance does not entirely preclude a weak saturation effect. With a baseline questionnaire of exactly $n = 75$ items, the statistical threshold for significance ($\alpha = 0.05$) requires an absolute correlation coefficient of roughly $|r| \geq 0.23$. Because our sample size creates a mathematical floor on detectability, the study is unequipped to definitively confirm or reject correlations weaker than this

threshold. However, the fact that all eight diverse operationalizations, consisting of maximum correlation, mean squared correlation, sum of squared correlations, and neighbor count, computed across both voters and candidates, yield slightly negative Spearman coefficients provides a weakly affirming signal of this saturation effect.

Ultimately, as will be shown in Section XI-C, pairwise correlations in the SmartVote questionnaire are structurally weak (mean $|\rho| = 0.190$, maximum $|\rho| = 0.775$). This native redundancy is simply too low to exert a dominant dampening effect on the recommendations, allowing raw answer variance to completely override correlation as the primary driver of adversarial impact.

Predictor	ρ_S	p-value
Voter answer variance	+0.806	< 0.001
Combined variance (voter $\times$ candidate)	+0.782	< 0.001
Candidate answer variance	+0.675	< 0.001
Voter skip rate (NaN %)	-0.654	< 0.001
Question position in survey	-0.556	< 0.001
Max $ \rho $ with other questions (voter)	-0.108	0.357
Max $ \rho $ with other questions (candidate)	-0.084	0.471
Mean r^2 with other questions (voter)	-0.148	0.205
Mean r^2 with other questions (candidate)	-0.086	0.463
Sum r^2 with other questions (voter)	-0.157	0.179
Sum r^2 with other questions (candidate)	-0.040	0.735
Number of correlated neighbours (voter)	-0.161	0.169
Number of correlated neighbours (candidate)	-0.051	0.662

Table VIII: Spearman rank correlations between question properties and clone impact ($1 - \bar{J}$ at $k = 4$ identical clones) across all 75 questions. Voter answer variance is the dominant predictor. None of the inter-question correlation metrics (below the separator) reach significance at $p < 0.05$. However, with $n = 75$ the study is powered to detect only effects of $|r| \gtrsim 0.23$, so weak relationships cannot be ruled out. The consistent non-significance across eight different operationalizations nonetheless suggests that inter-question correlation is at most a minor predictor of clone impact under L2 distance.

d) Consistency Across Robustness Measures.: To verify the robustness of these predictors, we evaluated whether the same question-level properties drive adversarial impact when measured at the party level (maximum absolute visibility change) rather than the recommendation level. The resulting hierarchy of predictors remains largely consistent, with two notable structural refinements (Table IX).

First, combined variance ($\rho_S = +0.621$) slightly overtakes voter variance ($\rho_S = +0.604$) as the strongest overall predictor. This is intuitively sound for a party-level metric: even if voters are highly polarized on an issue, if the political candidates themselves share a consensus position (low candidate variance), cloning the question will not meaningfully shift the relative visibility between competing parties.

Second, unlike the recommendation-level analysis, one inter-question correlation metric, namely mean voter r^2 , achieves weak statistical significance ($\rho_S = -0.244$, $p = 0.035$). This negative correlation provides empirical validation for the

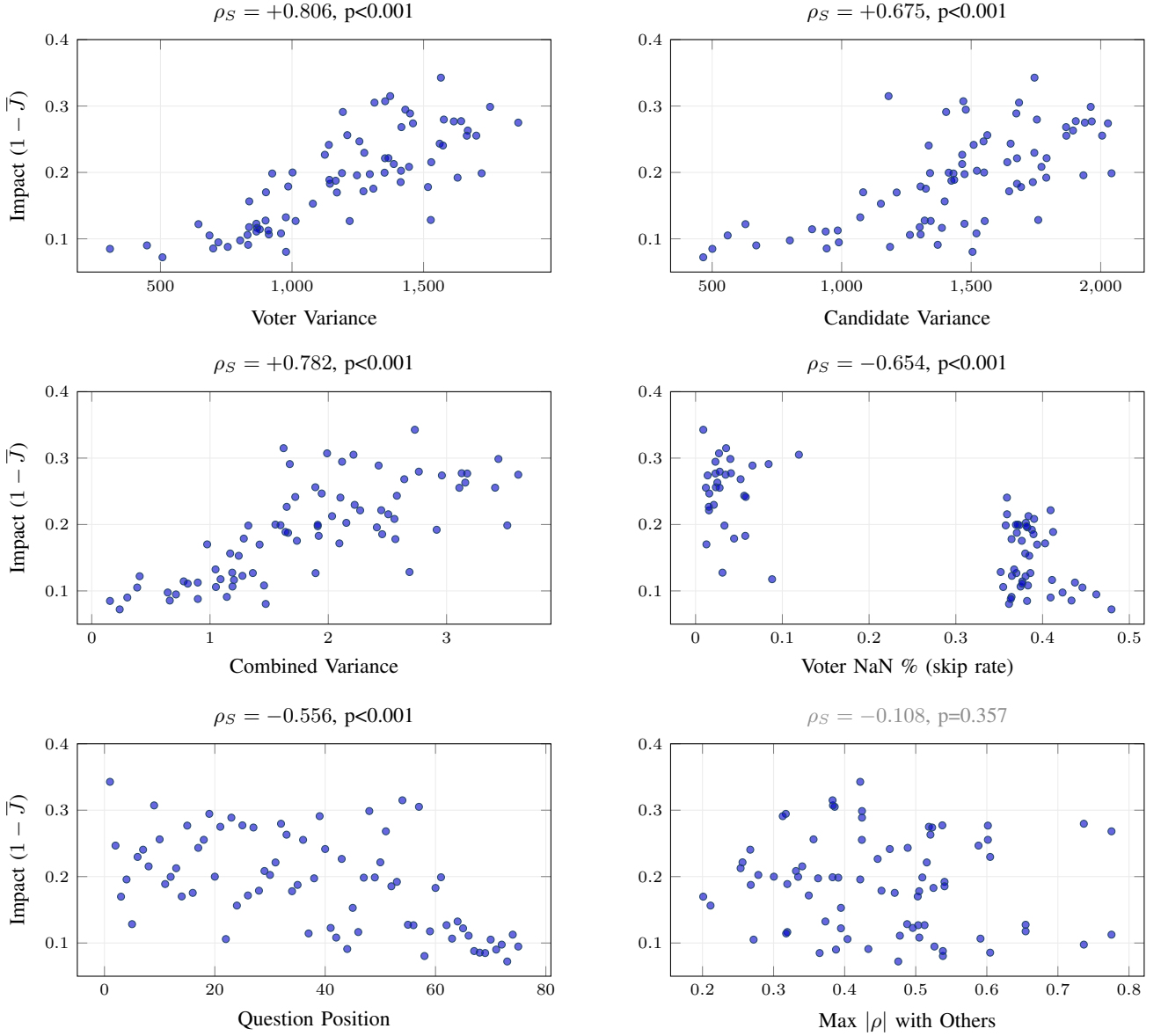


Figure 10: Predictors of per-question clone impact ($1 - \bar{J}$ at $k = 4$ identical clones). Each dot is one of 75 questions, titles show Spearman rank correlation (greyed out where $p \geq 0.05$). Combined variance is defined as the product of voter and candidate variance. Voter answer variance is the strongest predictor ($\rho_S = +0.806$): questions where voters spread across the answer spectrum produce larger squared differences under cloning. Maximum absolute correlation with other questions shows no significant relationship ($\rho_S = -0.108$, $p = 0.357$). However, with $n = 75$ the study is only powered to detect effects of $|r| \gtrsim 0.23$, so a weak relationship cannot be ruled out.

baseline saturation hypothesis discussed previously: questions whose underlying policy dimensions are already heavily represented in the questionnaire offer diminishing marginal returns for targeted party manipulation.

Finally, recommendation-level impact and party visibility impact are themselves only moderately correlated ($\rho_S = 0.688$), indicating that the questions causing the largest absolute recommendation shuffles are not necessarily the same questions yielding the highest concentrated partisan advantage.

B. Predictors of Embedding Model Performances

The model selection step in Section V identifies E5-Instruct as the best-performing model, but does not explain *why*. To understand which embedding properties drive CRW effectiveness, we construct a diagnostic benchmark from the 2019 SmartVote questionnaire (an independent dataset from the 2023 questionnaire used in all other experiments). We randomly select 14 questions, and for each of these anchor questions, we generate same-topic paraphrases and syn-

Predictor	ρ_S	p -value
Combined variance (voter $\times$ candidate)	+0.621	< 0.001
Voter answer variance	+0.604	< 0.001
Candidate answer variance	+0.598	< 0.001
Question position in survey	-0.576	< 0.001
Voter skip rate (NaN %)	-0.358	0.002
Mean r^2 with other questions (voter)	-0.244	0.035
Sum r^2 with other questions (voter)	-0.203	0.080
Max $ \rho $ with other questions (voter)	-0.182	0.119
Number of correlated neighbours (voter)	-0.168	0.149
Max $ \rho $ with other questions (candidate)	-0.152	0.193
Number of correlated neighbours (candidate)	-0.126	0.282
Mean r^2 with other questions (candidate)	-0.123	0.291
Sum r^2 with other questions (candidate)	-0.096	0.413

Table IX: Spearman rank correlations between question properties and party visibility impact ($\max_p |\Delta \nu_{36}^1(p)|$ at $k = 4$ identical clones) across all 75 questions. Combined variance is the product of voter and candidate variance. The predictor ordering mirrors Table VIII for direct comparison. Unlike the recommendation-level analysis, *mean* r^2 with other questions (voter) reaches significance at $p = 0.035$. However, with $n = 75$ the study is powered to detect only effects of $|r| \gtrsim 0.23$, so weaker relationships cannot be ruled out for the remaining metrics.

thaxic/topical trap questions using the GPT-4o.⁵ All generated pairs were manually inspected for quality.

To formalize these metrics, let q^* denote an anchor question. Let $\mathcal{S}$ be the set of same-topic variants (both paraphrases and negations), $\mathcal{N} \subset \mathcal{S}$ be the subset of strictly negated variants, and $\mathcal{T}$ be the set of trap variants. Let $d(x, y)$ represent the spatial embedding distance between two questions. For each anchor, the benchmark calculates the following three metrics (which are then averaged across all anchors):

- **Topic coherence (lower is better):** the ratio of the mean distance to same-topic variants over the mean distance to trap variants. Rather than measuring raw distance, this ratio captures whether a model successfully clusters functionally related questions tighter than deceptive, keyword-sharing traps. Formally:

$$\text{Topic Coherence}(q^*) = \frac{\frac{1}{|\mathcal{S}|} \sum_{s \in \mathcal{S}} d(q^*, s)}{\frac{1}{|\mathcal{T}|} \sum_{t \in \mathcal{T}} d(q^*, t)}$$

- **Negation invariance (lower is better):** the ratio of the mean distance to negated variants over the mean distance to trap variants. For CRW purposes, a question and its exact policy opposite (e.g., “Should immigration be increased?” and “Should immigration be restricted?”) address the identical functional dimension. The model must place them closer together than a keyword trap,

despite being semantically opposite. Formally:

$$\text{Negation Invariance}(q^*) = \frac{\frac{1}{|\mathcal{N}|} \sum_{n \in \mathcal{N}} d(q^*, n)}{\frac{1}{|\mathcal{T}|} \sum_{t \in \mathcal{T}} d(q^*, t)}$$

- **Ordering accuracy (higher is better):** the percentage of all possible combinations of (same-topic variant, trap variant) where the model correctly places the same-topic variant closer to the anchor than the trap. A score of 100% indicates perfect spatial separation between the valid policy dimension and the deceptive surface-level text. Formally:

$$\text{Ordering Accuracy}(q^*) = \frac{1}{|\mathcal{S}||\mathcal{T}|} \sum_{s \in \mathcal{S}} \sum_{t \in \mathcal{T}} \mathbb{I}(d(q^*, s) < d(q^*, t))$$

where $\mathbb{I}$ is the indicator function returning 1 if the distance condition is met and 0 otherwise.

Full benchmark details, including prompts, all 14 anchor pairs and the generate paraphrases and traps, are provided in Appendix XII.

a) *General Benchmark Performance.*: As detailed in Table X, the evaluated models exhibit stark differences in their ability to handle topical redundancy versus surface-level deception. The top-tier models, E5-Instruct and Jina v3, demonstrate exceptional topic coherence (0.368 and 0.409, respectively), indicating they successfully compress same-topic variants to roughly a third of the distance of deceptive keyword traps. Jina v3 is particularly robust to negations, achieving a near-perfect negation invariance score of 0.109. Conversely, models at the bottom of the ranking, such as E5-Asym and E5-Asym-Instruct, struggle significantly with semantic traps: their topic coherence scores approach 1.0 (0.808 and 0.883), meaning they place entirely unrelated questions sharing surface-level vocabulary almost as close together as genuine policy paraphrases.

Interestingly, global ordering accuracy remains relatively high across the board, with seven of the ten models exceeding 90%. E5-Asym-Instruct perfectly illustrates a critical disconnect here: despite its abysmal topic coherence, it still achieves 94.0% ordering accuracy.

b) *Topic coherence dominates:* Table X reports the performance of each model for each benchmark metric, while Table XI reports the Spearman correlation between each benchmark metric and CRW Jaccard performance across the 10 models. As visualized in Figure 11, topic coherence acts as an almost perfect predictor ($\rho_S = -0.915$, $p < 0.001$). Models that successfully form tight local clusters of same-topic variants relative to trap questions achieve dramatically better CRW performance. Negation invariance is the second-strongest predictor ($\rho_S = -0.733$, $p = 0.016$). Ordering accuracy, while positively associated, does not reach significance ($\rho_S = +0.624$, $p = 0.054$), indicating that correctly ordering global pairs matters less than compressing same-topic variants into a tight distance band.

⁵(gpt-4o-2024-08-06) with temperature 0.8 and structured JSON output.

Model	CRW J_{36}	Predict CRW		Do not predict CRW	
		Topic coh. ↓	Neg. inv. ↓	Ord. acc. ↑	Corr. ρ_S
E5-Instruct	0.585	0.368	0.198	99.4%	+0.105
Jina v3	0.581	0.409	0.109	99.7%	+0.192
GTE	0.563	0.604	0.247	90.2%	+0.166
Qwen3	0.565	0.459	0.146	98.5%	+0.198
BGE-M3	0.552	0.674	0.320	88.1%	+0.112
Nomic v2	0.542	0.631	0.220	93.2%	+0.128
E5	0.530	0.722	0.336	80.4%	+0.068
SBERT	0.541	0.587	0.134	92.3%	+0.060
E5-Asym	0.518	0.808	0.637	85.1%	+0.075
E5-Asym-Instruct	0.491	0.883	0.853	94.0%	+0.140

Table X: Embedding model performance at optimal α^* (Stage 1) alongside diagnostic benchmark metrics, sorted by CRW composite rank. Cell shading encodes each column’s rank independently: dark green = ranks 1–2 (best), light green = ranks 3–4, white = ranks 5–6, light red = ranks 7–8, dark red = ranks 9–10 (worst). The predictive metrics (topic coherence and negation invariance) track the CRW column closely, while ordering accuracy and correlation agreement show a more mixed pattern, reflecting their lack of predictive power (Table XI).

Benchmark metric	ρ_S	p -value	Significant
Topic coherence ↓	−0.915	< 0.001	Yes
Negation invariance ↓	−0.733	0.016	Yes
Ordering accuracy ↑	+0.624	0.054	No
Correlation agreement ρ_S	+0.455	0.187	No

Table XI: Spearman rank correlation between diagnostic benchmark metrics and CRW Jaccard performance (J_{36} at optimal α^*) across the 10 embedding models. Topic coherence is the strongest predictor of CRW effectiveness ($\rho_S = -0.915$). Correlation agreement with d_{corr} does not predict CRW performance.

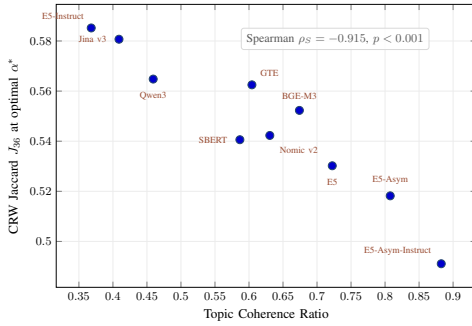


Figure 11: Topic coherence ratio vs. CRW Jaccard performance across all 10 models. The strong negative correlation ($\rho_S = -0.915$, $p < 0.001$) indicates that models which successfully cluster functionally related questions tighter than deceptive keyword traps achieve dramatically better CRW performance. E5-Asym-Instruct is a clear case of this: despite high global ordering accuracy (94%), its poor local topic coherence makes it the worst-performing CRW model.

c) Correlation agreement does not predict CRW performance.:

Table XI additionally reports each model’s Spearman agreement with the correlation-based distance matrix d_{corr} across all 2,775 question pairs. While all models exhibit a strictly positive rank correlation with empirical voter behavior, the absolute magnitude of this alignment is consistently weak, with individual model agreement scores ranging from $\rho_S = 0.060$ to $\rho_S = 0.198$. Figure 12 additionally visualizes this weak global alignment for E5-Instruct. Strikingly, a model’s baseline correlation agreement does not reliably predict its actual CRW effectiveness: when comparing these agreement scores against final CRW performance across the 10 models, the resulting meta-correlation is statistically insignificant ($\rho_S = +0.455$, $p = 0.187$). The top-performing model in baseline agreement, Qwen3, achieves a score of $\rho_S = 0.198$, while E5-Instruct (the definitively optimal model for CRW) ranks 7th with a mere $\rho_S = 0.105$.

E5-Asym-Instruct illustrates this dichotomy perfectly: it ranks 4th in global correlation agreement ($\rho_S = 0.140$) and achieves the 4th-highest global ordering accuracy (94.0%). Yet, it ranks last for CRW and possesses the worst local topic coherence (0.883) and negation invariance (0.853) of all 10 models, effectively failing to cluster related items tightly enough to trigger the defense.

C. The Global Distance Landscape

We analyze the pairwise distances across the 75-item SmartVote questionnaire ($\binom{75}{2} = 2,775$ unique pairs). As illustrated in Figures 13a and 14a, the global geometry is characterized by an empty left tail followed by a steep density ramp-up. Native survey questions almost never exhibit near-zero distances. The spatial distribution reveals a distinct lack of inherent clustering: rather than forming isolated components (or “policy islands”) separated by wide margins, the items populate a dense, highly interconnected space once the minimum pairwise distance threshold is crossed.

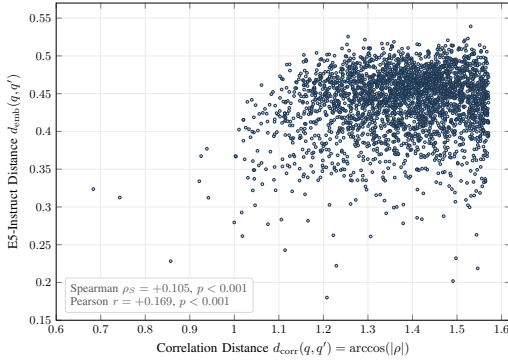


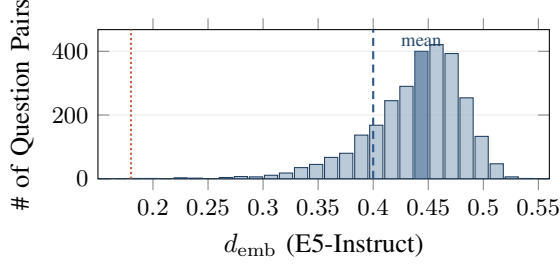
Figure 12: Relationship between correlation distance $d_{\text{corr}} = \arccos(|\rho|)$ and E5-Instruct embedding distance across all 2,775 question pairs. The weak positive correlation (Spearman $\rho_S = +0.105$, $p < 0.001$) confirms that textual similarity and functional redundancy are largely decoupled: E5-Instruct captures topical and lexical proximity, while correlation distance reflects voter response patterns that are not reliably predicted by question wording alone.

Figures 13b and 14b highlight this phenomenon by zooming in on the closest pairs. In the empirical correlation metric, there are only 8 pairs with a distance below $d_{\text{corr}} = 1.0$ ($|\rho| > 0.54$). However, as the distance increases even slightly beyond this barren tail, the cumulative number of question pairs skyrockets. Similarly, in the embedding space, only 23 pairs fall below $d_{\text{emb}} = 0.3$, but 96.8% of all pairs are densely packed in the narrow band between 0.3 and 0.5.

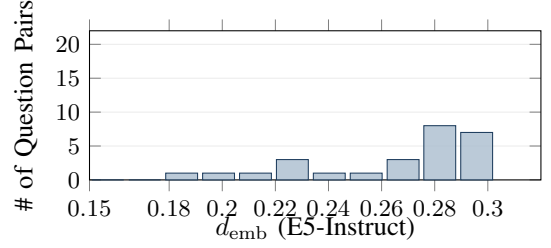
To understand the qualitative nature of these barren left tails, Tables XII and XIII detail the actual question pairs that populate the absolute extremes of both distributions. The divergence between the two metric spaces is stark: out of the top five closest pairs in each, only a single pair overlaps (foreign policy regarding the EU and Schengen agreements). This complete mismatch at the local extremes aligns with the weak global alignment ($\rho_S = 0.105$) established earlier in Section XI-B, confirming that even among the absolute nearest neighbors, embedding-space proximity and functional redundancy represent largely independent structures.

XII. IMPLEMENTATION DETAILS FOR MODEL EXPLAINABILITY ANALYSIS

All questions used in the embedding model benchmark, anchor questions drawn from the 2019 SmartVote questionnaire. Each group consists of one anchor question (shaded, bold) with its associated negation, easy and hard paraphrases, and negated variants, followed by four trap questions used to test embedding accuracy. *Syntax traps* share the same sentence structure as the anchor while *keyword traps* share surface vocabulary but address a different topic. All questions are in italics when they are traps. The exact prompts used are listed below.

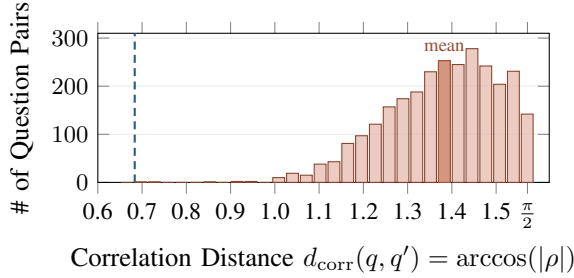


(a) The distribution is compressed into a narrow band: 96.8% of all pairs fall within $d_{\text{emb}} \in [0.3, 0.5]$, with a mean of 0.439 (darkened bar). The dashed line marks the optimal CRW radius $\alpha^* = 0.4$: 463 pairs (16.7%) fall within this radius. The dotted line marks the minimum observed distance (Q32215–Q32222: health insurance subsidies and health care deductibles, $d_{\min} = 0.180$).

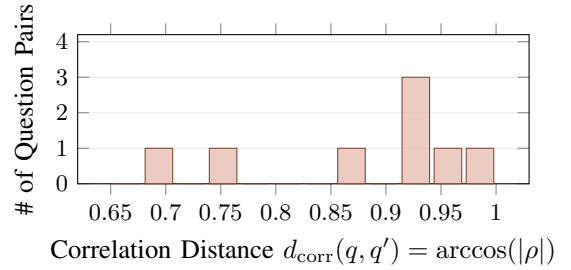


(b) Zoomed view of Figure 13a, showing all 23 question pairs with $d_{\text{emb}} < 0.3$ under E5-Instruct. The sparse left tail contains only 2 pairs below $d = 0.22$, the closest being Q32215–Q32222 (health insurance subsidies $\times$ health care deductibles, $d_{\min} = 0.180$). Even these closest pairs sit far from $d = 0$, confirming that no two questions in the clean questionnaire approach the near-zero distances that synthetic clones produce.

Figure 13: Distribution of pairwise E5-Instruct embedding distances across all $\binom{75}{2} = 2,775$ question pairs.



(a) The distribution is strongly right-skewed, with the density concentrated heavily in the upper range of the metric space ($1.3 \leq d \leq 1.5$). Only a small fraction of pairs fall below $d_{\text{corr}} = 1.0$ ($|\rho| > 0.54$), confirming that the SmartVote questionnaire contains very little functional redundancy as measured by voter response patterns. The vertical dashed line marks the minimum observed distance between any two questions, $d_{\min} = 0.68$ ($|\rho| = \cos(0.68) \approx 0.78$), c.f. Figure 14b. The highlighted bar indicates the bin containing the mean distance ($\bar{d} = 1.38$).



(b) Zoomed view of Figure 14a, showing all 8 question pairs with $d_{\text{corr}} < 1.0$ ($|\rho| > 0.54$). The two closest pairs—“army expansion” with “defense budget” ($d = 0.68$, $|\rho| = 0.78$) and “highway widening” with “road traffic budget” ($d = 0.74$, $|\rho| = 0.74$)—each occupy a bin alone, separated by a wide gap from the remaining 6 pairs clustered between $d = 0.86$ and $d = 1.0$. Even this closest cluster corresponds to only moderate correlations ($|\rho| \approx 0.54$ – 0.66), far below the $|\rho| \approx 1$ that identical or perfect clones produce.

Figure 14: Distribution of pairwise correlation distances $d_{\text{corr}}(q, q') = \arccos(|\rho(q, q')|)$ across all $\binom{75}{2} = 2,775$ question pairs.

Question q	Question q'	d_{emb}
Should the federal government allocate more funding for health insurance premium subsidies?	Should insured persons contribute more to health care costs (e.g., increase the minimum deductible)?	0.180
Should private households be free to choose their electricity supplier (complete liberalization of the electricity market)?	Would you be in favor of the introduction of increasing electricity tariffs when consumption is higher (progressive electricity tariffs)?	0.202
Should Switzerland strive for a comprehensive free trade agreement (including agriculture) with the USA?	Should Switzerland terminate the Bilateral Agreements with the EU and seek a free trade agreement without the free movement of persons?	0.219
Should the federal government spend more or less in the area of "Public transport"?	Should the federal government spend more or less in the area of "Road traffic (motorised individual transport)"?	0.222
Should Switzerland terminate the Schengen agreement with the EU and reintroduce more security checks directly on the border? [†]	Should Switzerland terminate the Bilateral Agreements with the EU and seek a free trade agreement without the free movement of persons? [†]	0.228

Table XII: The five question pairs with smallest E5-Instruct embedding distance. The shaded row (†) is the only pair appearing in both top-5 lists.

Question q	Question q'	$ \rho $	d_{corr}
Are you in favor of expanding the army's target number of soldiers to at least 120,000?	Should the federal government spend more or less in the area of "National defence"?	0.775	0.683
Should busy sections of highways be widened?	Should the federal government spend more or less in the area of "Road traffic (motorised individual transport)"?	0.737	0.743
Should Switzerland terminate the Schengen agreement with the EU and reintroduce more security checks directly on the border? [†]	Should Switzerland terminate the Bilateral Agreements with the EU and seek a free trade agreement without the free movement of persons? [†]	0.655	0.857
Should the federal government provide more financial support for public housing construction?	Should the federal government spend more or less in the area of "Social services"?	0.605	0.921
Should the conditions for naturalization be relaxed (e.g., shorter residence period)?	Should foreign nationals who have lived in Switzerland for at least ten years be granted the right to vote and stand for election at the municipal level?	0.602	0.925

Table XIII: The five question pairs with smallest correlation distance $d_{\text{corr}} = \arccos(|\rho|)$, i.e. the most functionally redundant pairs in the questionnaire as measured by voter and candidate response patterns. The shaded row (†) also appears in Table XII: the only pair in both top-5 lists.

Type	Question
Anchor	Should cannabis use be legalized?
Negation	Should cannabis use not be legalized?
Easy par.	Is the legalization of cannabis use a good idea?
Easy par. neg.	Is it a bad idea to legalize cannabis use?
Hard par.	Is it appropriate to allow the consumption of marijuana according to the law?
Hard par.	Ought the law to permit individuals to use marijuana freely?
Hard par. neg.	The law should not permit individuals to consume marijuana.
<i>Syntax trap</i>	<i>Should income taxes be reduced?</i>
<i>Syntax trap</i>	<i>Should public transportation fares be increased?</i>
<i>Keyword trap</i>	<i>Is cannabis a widely used plant in gardening?</i>
<i>Keyword trap</i>	<i>Is the use of legal documents important in business transactions?</i>
Anchor	Should Switzerland start membership negotiations with the EU?
Negation	Should Switzerland not start membership negotiations with the EU?
Easy par.	Is it time for Switzerland to begin talks about joining the European Union?
Easy par. neg.	Is it time for Switzerland to refrain from beginning talks about joining the European Union?
Hard par.	Is it appropriate for Switzerland to commence discussions to become a part of the European Union?
Hard par.	Should Switzerland engage in discussions to potentially unify with the European Union?
Hard par. neg.	It is not advisable for Switzerland to engage in discussions to potentially unify with the European Union.
<i>Syntax trap</i>	<i>Should Switzerland implement a new national healthcare policy?</i>
<i>Syntax trap</i>	<i>Should Switzerland adopt new educational standards for primary schools?</i>
<i>Keyword trap</i>	<i>Should negotiations start for Switzerland's participation in international sports events?</i>
<i>Keyword trap</i>	<i>Is Switzerland ready to begin membership discussions for a global technology consortium?</i>
Anchor	Should the conversion rate of the occupational pension fund be reduced in order to adjust for increases in life expectancy?
Negation	Should the conversion rate of the occupational pension fund not be reduced in order to adjust for increases in life expectancy?
Easy par.	Is it advisable to decrease the conversion rate of occupational pension funds to accommodate longer life expectancy?
Easy par. neg.	Is it advisable to keep the conversion rate of occupational pension funds unchanged despite longer life expectancy?
Hard par.	Is it necessary to lower the pension conversion factor to reflect the rise in average lifespan?
Hard par.	Should adjustments be made to the retirement fund rate to account for people living longer on average?
Hard par. neg.	Maintaining the current pension conversion rate should continue, even as people live longer.
<i>Syntax trap</i>	<i>Should the carbon emission targets for industrial sectors be increased to combat climate change?</i>
<i>Syntax trap</i>	<i>Should the funding for public education be enhanced to improve literacy rates?</i>
<i>Keyword trap</i>	<i>Should the conversion rate for reward points in airline programs be reduced to adjust for increased travel demand?</i>
<i>Keyword trap</i>	<i>Should the conversion rate of foreign currency exchanges be adjusted in response to market fluctuations?</i>

Continued on next page

Table XIV continued

Type	Question
Anchor	Should private households be free to choose their electricity supplier (complete liberalisation of the electricity market)?
Negation	Should private households not be free to choose their electricity supplier (complete liberalisation of the electricity market)?
Easy par.	Ought individual households have the liberty to select their own electricity provider, allowing full liberalisation of the electricity market?
Easy par. neg.	Should individual households not have the freedom to select their own electricity provider, thus preventing full liberalisation of the electricity market?
Hard par.	Is it necessary for households to have the autonomy to pick their own energy provider, leading to a fully deregulated power market?
Hard par.	Do families require the ability to independently choose who supplies their power, resulting in an entirely open electricity sector?
Hard par. neg.	Households must be restricted from independently choosing their energy provider to prevent the energy market from becoming completely deregulated.
<i>Syntax trap</i>	<i>Should public schools be free to choose their own curriculum (complete decentralisation of educational policy)?</i>
<i>Syntax trap</i>	<i>Should private companies be free to set their own environmental standards (complete deregulation of environmental policies)?</i>
<i>Keyword trap</i>	<i>Should private households be free to choose their internet provider for faster service?</i>
<i>Keyword trap</i>	<i>Should private households be able to select their water supplier for better quality?</i>
Anchor	What is your position the following statement: “The ongoing digitalization offers significantly more opportunities than risks.”
Negation	What is your position on the following statement: “The ongoing digitalization does not offer significantly more opportunities than risks.”
Easy par.	How do you feel about the statement that digitalization currently presents many more opportunities than it does risks?
Easy par. neg.	How do you feel about the statement that digitalization currently presents many more risks than it does opportunities?
Hard par.	Do you agree that the transition to digital technologies provides far greater advantages compared to the potential downsides?
Hard par.	Would you say that the benefits of moving towards a digital age outweigh the associated challenges?
Hard par. neg.	Do you believe that the transition to digital technologies results in more disadvantages than benefits?
<i>Syntax trap</i>	<i>What is your position on the following statement: “The reduction in military spending poses significantly more threats than benefits.”</i>
<i>Syntax trap</i>	<i>What is your position on the following statement: “Introducing universal healthcare offers significantly more costs than savings.”</i>
<i>Keyword trap</i>	<i>What is your position on the following statement: “The ongoing digitalization in art offers significantly more inspiration than limitations.”</i>
<i>Keyword trap</i>	<i>What is your position on the following statement: “The ongoing digitalization of culinary techniques offers significantly more creativity than constraints.”</i>
Anchor	Should the rules for reproductive medicine be further relaxed?
Negation	Should the rules for reproductive medicine not be further relaxed?
Easy par.	Is it necessary to loosen the regulations governing reproductive medicine?
Easy par. neg.	Is it necessary to tighten the regulations governing reproductive medicine?
Hard par.	Is there a need to make the guidelines for assisted reproductive techniques more lenient?

Continued on next page

Table XIV continued

Type	Question
Hard par.	Should the constraints on technologies for human reproduction be eased?
Hard par. neg.	It is essential to maintain or even tighten the current restrictions on medical procedures related to reproduction.
<i>Syntax trap</i>	<i>Should the standards for educational curricula be further revised?</i>
<i>Syntax trap</i>	<i>Should the policies regarding renewable energy be further developed?</i>
<i>Keyword trap</i>	<i>Should the rules for laboratory methods be further relaxed in scientific research?</i>
<i>Keyword trap</i>	<i>Should the regulations for food production be further relaxed to enhance culinary diversity?</i>
Anchor	Should the federal government provide more support for the construction of non-profit housing?
Negation	Should the federal government not provide more support for the construction of non-profit housing?
Easy par.	Is it necessary for the national government to increase its support for building housing that is not for profit?
Easy par. neg.	Is it unnecessary for the national government to increase its support for building housing that is not for profit?
Hard par.	Is there a need for the central authorities to enhance their financial assistance for developing housing projects with no profit motive?
Hard par.	Ought the national administration to boost aid for initiatives aimed at creating housing that lacks a profit-driven focus?
Hard par. neg.	The central authorities should refrain from enhancing financial aid for developing projects intended for housing without a profit goal.
<i>Syntax trap</i>	<i>Should the federal government provide more support for the development of renewable energy sources?</i>
<i>Syntax trap</i>	<i>Should the federal government provide more support for the improvement of public transportation systems?</i>
<i>Keyword trap</i>	<i>Should private organizations receive more support for the construction of new office buildings?</i>
<i>Keyword trap</i>	<i>Should the federal government provide more support for the construction of sports facilities?</i>
Anchor	Should the federal government spend more or less in the area of “Public security”?
Negation	Should the federal government spend less or more in the area of “Public security”?
Easy par.	Should the federal government increase or decrease its spending on public security?
Easy par. neg.	Should the federal government decrease or increase its spending on public security?
Hard par.	Is it advisable for the national authorities to allocate additional resources towards ensuring public safety?
Hard par.	Should national funding be adjusted to provide more emphasis on the protection of citizens?
Hard par. neg.	The national authorities ought to reduce the allocation of resources towards ensuring public safety.
<i>Syntax trap</i>	<i>Should the federal government spend more or less in the area of “Environmental protection”?</i>
<i>Syntax trap</i>	<i>Should the federal government spend more or less in the area of “Education reform”?</i>
<i>Keyword trap</i>	<i>Should the government spend more on public holidays?</i>

Continued on next page

Table XIV continued

Type	Question
<i>Keyword trap</i>	<i>Should more public funds be allocated to organizing community events?</i>
Anchor	Should Switzerland strive for a free trade agreement with the USA?
Negation	Should Switzerland not strive for a free trade agreement with the USA?
Easy par.	Is it advisable for Switzerland to pursue a free trade deal with the United States?
Easy par. neg.	Is it advisable for Switzerland not to pursue a free trade deal with the United States?
Hard par.	Would it be beneficial for Switzerland to establish a commercial treaty with the United States?
Hard par.	Should Switzerland work towards forming a bilateral trade pact with America?
Hard par. neg.	It would not be in Switzerland's interest to establish a commercial treaty with the United States.
<i>Syntax trap</i>	<i>Should Switzerland invest more resources in expanding its public transportation network?</i>
<i>Syntax trap</i>	<i>Should Switzerland implement stricter regulations on digital privacy?</i>
<i>Keyword trap</i>	<i>Should Switzerland strive to maintain its position in international sporting events with the USA?</i>
<i>Keyword trap</i>	<i>Should Switzerland form a cultural exchange program with the USA?</i>
Anchor	Are you in favour of restricting competition between the cantons with regard to corporate tax rates?
Negation	Are you in favour of not restricting competition between the cantons with regard to corporate tax rates?
Easy par.	Do you support limiting the competition between cantons in terms of their corporate tax rates?
Easy par. neg.	Do you oppose limiting the competition between cantons in terms of their corporate tax rates?
Hard par.	Do you agree with the idea that cantons should not undercut each other by offering lower corporate taxes?
Hard par.	Should there be a ban on the practice of cantons engaging in a race to the bottom with corporate tax incentives?
Hard par. neg.	Cantons should have the freedom to engage in competitive tax rate setting without restrictions.
<i>Syntax trap</i>	<i>Are you in favour of enforcing stricter regulations on the import of non-native animal species?</i>
<i>Syntax trap</i>	<i>Are you in favour of increasing government funding for renewable energy research and development?</i>
<i>Keyword trap</i>	<i>Are you in favour of restricting access to competitive sports events between different cantons?</i>
<i>Keyword trap</i>	<i>Are you in favour of imposing limits on the number of corporate events a canton can host?</i>
Anchor	Should online brokerage services (e.g. "Airbnb" accommodations, "Uber" taxi services) be regulated more strongly?
Negation	Should online brokerage services (e.g. "Airbnb" accommodations, "Uber" taxi services) not be regulated more strongly?
Easy par.	Is there a need for stricter regulation of online platforms such as Airbnb and Uber?
Easy par. neg.	Is there no need for stricter regulation of online platforms such as Airbnb and Uber?
Hard par.	Do digital service platforms like those for lodging and ride-sharing require tighter governmental oversight?

Continued on next page

Table XIV continued

Type	Question
Hard par.	Ought platforms providing digital accommodations and transport services face increased regulatory control?
Hard par. neg.	Digital services for accommodation and transportation do not require any additional regulatory measures.
Syntax trap	<i>Should agricultural subsidies (e.g. “corn” production, “dairy” farming) be reduced significantly?</i>
Syntax trap	<i>Should educational curriculums (e.g. “mathematics” courses, “science” labs) be updated frequently?</i>
Keyword trap	<i>Should Airbnb and Uber collaborate on a new online marketing strategy?</i>
Keyword trap	<i>Is there a more efficient way for Uber and Airbnb to expand their customer base?</i>
Anchor	Should the current moratorium on genetically modified plants and animals in Swiss agriculture be extended beyond 2021?
Negation	Should the current moratorium on genetically modified plants and animals in Swiss agriculture not be extended beyond 2021?
Easy par.	Ought the existing ban on genetically modified plants and animals in Swiss farming be continued past 2021?
Easy par. neg.	Ought the existing ban on genetically modified plants and animals in Swiss farming not be continued past 2021?
Hard par.	Is it advisable to prolong the prohibition against genetically engineered flora and fauna in Switzerland’s agricultural sector beyond the year 2021?
Hard par.	Should Switzerland maintain its restriction on the cultivation of genetically altered crops and livestock after 2021?
Hard par. neg.	Is it advisable to lift the prohibition against genetically engineered flora and fauna in Switzerland’s agricultural sector after 2021?
Syntax trap	<i>Should the current restrictions on foreign investments in Swiss real estate be extended beyond 2021?</i>
Syntax trap	<i>Should the existing limits on digital surveillance in Switzerland be extended beyond 2021?</i>
Keyword trap	<i>Should the current moratorium on genetically modified foods in international markets be extended beyond 2021?</i>
Keyword trap	<i>Should the current moratorium on genetically engineered animals in medical research be extended beyond 2021?</i>
Anchor	Should the government increase its efforts to support equal education opportunities (e.g. through vouchers for private tutoring for students from low-income families)?
Negation	Should the government not increase its efforts to support equal education opportunities (e.g. through vouchers for private tutoring for students from low-income families)?
Easy par.	Is it important for the government to enhance its initiatives in promoting equal access to educational opportunities, such as providing vouchers for private tutoring to students from families with lower incomes?
Easy par. neg.	Isn’t it unnecessary for the government to enhance its initiatives in promoting equal access to educational opportunities, such as providing vouchers for private tutoring to students from families with lower incomes?
Hard par.	Ought the state to amplify its role in ensuring all students have the same educational chances, possibly by offering financial aid for private lessons to those from financially disadvantaged backgrounds?
Hard par.	Should public authorities boost their efforts to level the playing field in education, perhaps by granting subsidies for additional academic support to pupils from economically challenged households?

Continued on next page

Table XIV continued

Type	Question
Hard par. neg.	The state should refrain from amplifying its role in ensuring equal educational opportunities, such as through financial aid for private lessons for students from financially disadvantaged backgrounds.
Syntax trap	<i>Should the government increase its efforts to support renewable energy sources (e.g. through subsidies for solar panel installations in rural areas)?</i>
Syntax trap	<i>Should the government increase its efforts to improve public transportation infrastructure (e.g. through funding for new railway lines in underserved regions)?</i>
Keyword trap	<i>Should the government increase its efforts to support equal employment opportunities (e.g. through internships for individuals from diverse backgrounds)?</i>
Keyword trap	<i>Should the government increase its efforts to support equal healthcare access (e.g. through subsidies for medical treatments for low-income families)?</i>
Anchor	An initiative calls for the introduction of paid paternity leave for four weeks. Do you support this proposal?
Negation	An initiative calls for the introduction of unpaid paternity leave for four weeks. Do you support this proposal?
Easy par.	Do you agree with the initiative proposing a four-week period of paid leave for new fathers?
Easy par. neg.	Do you oppose the initiative proposing a four-week period of paid leave for new fathers?
Hard par.	Should a proposal be enacted to allow fathers to receive salary compensation for a month after the birth of a child?
Hard par.	Is it advisable for a policy to be introduced granting one month of compensated time off for fathers following the arrival of a newborn?
Hard par. neg.	Introducing a policy that grants fathers a month of paid leave after a child is born is not something that deserves support.
Syntax trap	<i>An initiative calls for the introduction of a new carbon tax to combat climate change. Do you support this proposal?</i>
Syntax trap	<i>An initiative calls for the introduction of stricter regulations on data privacy. Do you support this proposal?</i>
Keyword trap	<i>An initiative calls for the introduction of a four-week training program for new employees. Do you support this proposal?</i>
Keyword trap	<i>An initiative calls for the introduction of paid parking for four weeks during the holiday season. Do you support this proposal?</i>

You are helping generate test data for evaluating embedding models on a Swiss political questionnaire. We need variants of a question that test whether models understand TOPIC similarity (not just word overlap).

Original question:

"{question_text}"

Generate ALL of the following variants. Return a JSON object with these exact keys:

- 1) "negation": Negate the original using the simplest change. Insert *not* or swap a key word for its antonym (e.g. *increase* → *decrease*). Keep everything else identical. Someone who agrees with the original should disagree with your version.
- 2) "easy_paraphrase": Rephrase using similar but not identical wording and slightly different structure. The meaning must be exactly the same. Some word overlap with the original is fine.
- 3) "easy_paraphrase_negation": Create a paraphrase (similar wording, slightly different structure) and then negate it. Someone who agrees with the original should disagree. Return only the final negated paraphrase.
- 4) "hard_paraphrase_1": Rephrase using COMPLETELY DIFFERENT words and sentence structure. The meaning must be identical, but you MUST NOT share significant content words with the original OR with the *easy_paraphrase* above. Use synonyms, different grammatical forms, or circumlocutions. You may change question to statement or vice versa.
- 5) "hard_paraphrase_2": Another hard paraphrase, different from *hard_paraphrase_1*. Again, NO shared content words with the original. Must also differ from *hard_paraphrase_1*.
- 6) "hard_paraphrase_negation": Create a hard paraphrase (completely different words) and negate it. Must clearly express the OPPOSITE stance. Not just a rhetorical question — an unambiguous disagreement with the original.
- 7) "syntax_trap_1": A question about a COMPLETELY DIFFERENT political topic that mimics the syntactic structure of the original. Must be about a genuinely unrelated policy area.
- 8) "syntax_trap_2": Another syntax trap, different topic from *syntax_trap_1*. Again, unrelated policy area.
- 9) "keyword_trap_1": A sentence using keywords from the original but about a COMPLETELY DIFFERENT topic — either a non-political context or an unrelated policy area. Should fool a naive keyword-matcher but not a good topic model.
- 10) "keyword_trap_2": Another keyword trap, different from *keyword_trap_1*.

IMPORTANT RULES:

- Every variant must be a single sentence or question.
- Hard paraphrases MUST use different vocabulary from the original AND from the easy paraphrase. If the original says *taxes*, use *fiscal levies* or *government revenue*. If it says *legalized*, use *permitted by law* or *decriminalized*.
- All negations must UNAMBIGUOUSLY express the opposite stance. Not a double negative or rhetorical question.
- Traps must be about genuinely different topics — not the same topic with a slight twist.
- Each variant must be distinct from all others.

Return valid JSON with exactly these 10 keys.

Figure 15: Exact GPT-4o prompt (gpt-4o-2024-08-06, temperature 0.8, structured JSON output) used to generate all ten benchmark variants for each anchor question. The placeholder {question_text} is replaced with the anchor text at call time.

XIII. FULL QUESTIONNAIRE

All 75 questions in the 2023 SmartVote questionnaire for the Swiss National Council elections, grouped by topic category. *Options* indicates the number of answer choices (4 = policy, 5 = budget, 7 = value). *Mini* indicates inclusion in the 30-question *rapide* questionnaire (✓).

ID	Question	Category	Opt.	Mini
Q32258	Should the Swiss mobile network be equipped throughout the country with the latest technology (currently 5G standard)?	Democracy & Media	4	
Q32259	Should the federal government be given additional powers in the area of digitization of government services in order to be able to impose binding directives and standards on the cantons?	Democracy & Media	4	
Q32260	Are you in favor of stronger regulation of the major Internet platforms (i.e., transparency rules on algorithms, increased liability for content, combating disinformation)?	Democracy & Media	4	✓
Q32261	A popular initiative aims to reduce television and radio fees (CHF 200 per household, exemption for businesses). Do you support this initiative?	Democracy & Media	4	✓
Q32262	Are you in favour of lowering the voting age to 16?	Democracy & Media	4	
Q32263	Should it be possible to hold a referendum on federal spending above a certain amount (optional financial referendum)?	Democracy & Media	4	
Q32240	Are you in favor of introducing a minimum wage of CHF 4,000 for all full-time employees?	Economy & Labour	4	✓
Q32241	Do you support stricter regulations for the financial sector (e.g., stricter capital requirements for banks, ban on bonuses)?	Economy & Labour	4	✓
Q32242	Should private households be free to choose their electricity supplier (complete liberalization of the electricity market)?	Economy & Labour	4	
Q32243	Should housing construction regulations be relaxed (e.g., noise protection, occupancy rates)?	Economy & Labour	4	
Q32244	Are you in favor of stricter controls on equal pay for women and men?	Economy & Labour	4	✓
Q32225	According to the Swiss integrated schooling concept, children with learning difficulties or disabilities should be taught in regular classes. Do you approve of this concept?	Education	4	
Q32226	Should the federal government raise the requirements for the gymnasiale matura?	Education	4	
Q32227	Should the state be more committed to equal educational opportunities (e.g., through subsidized remedial courses for students from low-income families)?	Education	4	✓
Q32245	Should busy sections of highways be widened?	Energy & Transport	4	✓
Q32246	Should Switzerland ban the registration of new passenger cars with internal combustion engines starting in 2035?	Energy & Transport	4	✓
Q32247	To achieve climate targets, should incentives and target agreements be relied on exclusively, rather than bans and restrictions?	Energy & Transport	4	
Q32248	Do you think it's fair that environmental and landscape protection rules are being relaxed to allow for the development of renewable energy?	Energy & Transport	4	
Q32249	Should the construction of new nuclear power plants in Switzerland be allowed again?	Energy & Transport	4	✓
Q32250	Should the state guarantee a comprehensive public service offering also in rural regions?	Energy & Transport	4	

Continued on next page

Table XV continued

ID	Question	Category	Opt.	Mini
Q32251	Would you be in favor of the introduction of increasing electricity tariffs when consumption is higher (progressive electricity tariffs)?	Energy & Transport	4	
Q32281	Should the federal government spend more or less in the area of "Social services"?	Federal Budget	5	
Q32282	Should the federal government spend more or less in the area of "Education and research"?	Federal Budget	5	
Q32283	Should the federal government spend more or less in the area of "Agriculture"?	Federal Budget	5	
Q32284	Should the federal government spend more or less in the area of "Public transport"?	Federal Budget	5	
Q32285	Should the federal government spend more or less in the area of "Road traffic (motorised individual transport)"?	Federal Budget	5	
Q32286	Should the federal government spend more or less in the area of "Public security"?	Federal Budget	5	
Q32287	Should the federal government spend more or less in the area of "National defence"?	Federal Budget	5	
Q32288	Should the federal government spend more or less in the area of "Development assistance"?	Federal Budget	5	
Q32236	Do you support tax cuts at the federal level over the next four years?	Finances & Taxes	4	✓
Q32237	Should married couples be taxed separately (individual taxation)? [BePart question]	Finances & Taxes	4	
Q32238	Would you support the introduction of a national inheritance tax on all inheritances over one million Swiss francs? [BePart question]	Finances & Taxes	4	✓
Q32239	Should the differences between cantons with high and low financial capacity be further reduced through fiscal equalization?	Finances & Taxes	4	
Q32269	Are you in favor of closer relations with the European Union (EU)?	Foreign Policy	4	
Q32270	Should Switzerland strive for a comprehensive free trade agreement (including agriculture) with the USA?	Foreign Policy	4	✓
Q32271	Should Swiss companies be obliged to ensure that their subsidiaries and suppliers operating abroad comply with social and environmental standards?	Foreign Policy	4	
Q32272	Should Switzerland terminate the Bilateral Agreements with the EU and seek a free trade agreement without the free movement of persons?	Foreign Policy	4	✓
Q32273	Should Switzerland return to a strict interpretation of neutrality (renounce economic sanctions to a large extent)? [BePart question]	Foreign Policy	4	✓
Q32220	Should compulsory vaccination of children be introduced based on the Swiss vaccination plan?	Health	4	
Q32221	Are you in favor of the introduction of a tax on foods containing sugar (sugar tax)?	Health	4	
Q32222	Should insured persons contribute more to health care costs (e.g., increase the minimum deductible)?	Health	4	✓
Q32223	Should the Federal Council's ability to restrict private and economic life in the event of a pandemic be more limited?	Health	4	✓

Continued on next page

Table XV continued

ID	Question	Category	Opt.	Mini
Q32224	Should the federal government be given the authority to determine the hospital offering (national hospital planning with regard to locations and range of services)?	Health	4	
Q32228	Should the conditions for naturalization be relaxed (e.g., shorter residence period)?	Immigration	4	✓
Q32229	Should more qualified workers from non-EU/EFTA countries be allowed to work in Switzerland (increase third-country quota)?	Immigration	4	
Q32230	Do you support efforts to house asylum seekers in centers outside Europe during the asylum procedure?	Immigration	4	✓
Q32231	Should foreign nationals who have lived in Switzerland for at least ten years be granted the right to vote and stand for election at the municipal level?	Immigration	4	✓
Q32252	Are you in favor of further relaxing the protection regulations for large predators (lynx, wolf, bear)?	Nature Conservation	4	✓
Q32253	Should direct payments only be granted to farmers with proof of ecological performance?	Nature Conservation	4	✓
Q32254	Are you in favour of stricter animal welfare regulations for livestock (e.g. permanent access to outdoor areas)?	Nature Conservation	4	
Q32255	Should 30% of Switzerland's land area be dedicated to preserving biodiversity? [BePart question]	Nature Conservation	4	
Q32256	Would you support a ban on single-use plastic and non-recyclable plastics? [BePart question]	Nature Conservation	4	✓
Q32257	Are you in favour of government measures to make the use of electronic devices more sustainable (e.g., right to repair, extension of warranty period, minimum guaranteed period for software updates)? [BePart question]	Nature Conservation	4	
Q32264	Are you in favor of expanding the army's target number of soldiers to at least 120,000?	Security & Military	4	✓
Q32265	Should the Swiss Armed Forces expand their cooperation with NATO?	Security & Military	4	
Q32266	Should the Federal Council be allowed to authorize other states to re-export Swiss weapons in cases of a war of aggression in violation of international law (e.g., the attack on Ukraine)? [BePart question]	Security & Military	4	
Q32267	Should automatic facial recognition be banned in public spaces?	Security & Military	4	✓
Q32268	Should Switzerland terminate the Schengen agreement with the EU and reintroduce more security checks directly on the border? [BePart question]	Security & Military	4	✓
Q32232	Should cannabis use be legalized? [BePart question]	Society & Ethics	4	✓
Q32233	Would you be in favour of doctors being allowed to administer direct active euthanasia in Switzerland?	Society & Ethics	4	
Q32234	Should a third official gender be introduced alongside "female" and "male"?	Society & Ethics	4	✓
Q32235	Do you think it's right for same-sex couples to have the same rights as heterosexual couples in all areas?	Society & Ethics	4	
Q32274	What is your position the following statement: "Someone who is not guilty, has nothing to fear from state security measures."	Values	7	

Continued on next page

Table XV continued

ID	Question	Category	Opt.	Mini
Q32275	What is your position the following statement: "In the long term, everyone benefits from a free market economy in the long term."	Values	7	
Q32276	What is your position the following statement: "It is necessary for the State to balance out differences in income and wealth through redistribution".	Values	7	
Q32277	What is your position the following statement: "It is best for a child, when one parent stays home full-time for childcare."	Values	7	
Q32278	What is your position the following statement: "The on-going digitalization offers significantly more opportunities than risks."	Values	7	
Q32279	What is your position the following statement: "Punishing criminals is more important than reintegrating them into society."	Values	7	
Q32280	What is your position the following statement: "Stronger environmental protection is necessary, even if its application limits economic growth."	Values	7	
Q32214	Do you support an increase in the retirement age (e.g., to 67)?	Welfare & Family	4	✓
Q32215	Should the federal government allocate more funding for health insurance premium subsidies?	Welfare & Family	4	✓
Q32216	For married couples, the pension is currently limited to 150% of the maximum individual AHV pension (capping). Should this limit be eliminated?	Welfare & Family	4	
Q32217	As part of the reform of the BVG (occupational pension plan), pensions are to be reduced (lowering the minimum conversion rate from 6.8% to 6%). Are you in favor of this measure?	Welfare & Family	4	
Q32218	Should paid parental leave be increased beyond today's 14 weeks of maternity leave and two weeks of paternity leave? [BePart question]	Welfare & Family	4	
Q32219	Should the federal government provide more financial support for public housing construction? [BePart question]	Welfare & Family	4	✓